

Behavioral Quotients as a Design Principle for Computation: Geometry, Compression, and Commitment

Unification, one geometric construction, externally verified implementations, and a research agenda

Status note. Every structural claim below is placed against prior art. The quotient/minimization theorem is the Myhill-Nerode congruence [1,2], its stochastic form is causal-state minimality [6,7,8], its MDP form is exact state aggregation [9,10], and its categorical form is the coequalizer / final-coalgebra universal property [13,14,19]. The generated geometry is the Gromov visual metric on a tree boundary [24,25] and the Fisher-Rao geometry of a statistical manifold [31,32,33,34]. The genuinely novel parts are of unequal status: prime weighting (§5.2) and the Riemann “asymptotic lock” (§5.3) remain **conjectural**, while the solver line’s correctness results are **externally verified self-evaluation** — witnesses checked against the original instances by separate verifiers [154,155,156,157,158] — even though the Laplace-horizon derivation remains informal. Provenance determines independence; verification determines validity (§7.5).

Contribution. The unifying theorem is not new (Appendix D): it is the universal property of a quotient, i.e. the coequalizer, and it coincides definitionally with Myhill-Nerode, causal-state, bisimulation, and minimal-realization theory. What this paper offers is a common lens and a design criterion, not a new minimality theorem. Appendix D marks every cross-regime bridge as a *consequence*, an *analogy*, or a *conjecture*, and states what would upgrade it. The paper is integrated by design: the solver line (§6.2, §7.3, §7.5) is a reference implementation of the design criterion, and its correctness results are externally verified. §0 states exactly which evidence tier each claim occupies, so that the theory does not borrow the solver’s verification and the solver does not borrow the theory’s novelty.

Abstract

Computational systems face a single tension: retaining exhaustive history grows state without bound and paralyzes search, while discarding it prematurely causes irrecoverable commitment errors [6,7,42,45]. We formalize state reduction as a **quotient of histories by future indistinguishability**. We show the minimal sufficient state is that quotient and that it satisfies the universal factorization property of a coequalizer [19]; in the finite deterministic case this is the Myhill-Nerode theorem [1,2], in the stochastic case the causal states of computational mechanics [6,7], and in the MDP case exact state aggregation [9,10]. Geometry is then not imposed *a priori* but generated by distinguishability depth: projective limits of observation partitions yield ultrametric tree boundaries [24,25,26], whose continuous relaxation is the Fisher-Rao/natural-gradient geometry of the uncertainty simplex [31,32,34,35] together with a hyperbolic cusp completion [25,36]. Constraint compatibility is a presheaf local-to-global problem with Čech cohomological obstructions [37,38,39]. Thermodynamic relaxation under log-barrier and prime-weighted potentials is Lyapunov-stable [47,48,59,60]; multiplicative (log-domain) gating preserves gradient direction where additive penalties cancel [52,53]. Four instantiations follow: KV-cache quotienting [83,84,85,130,131], answer-preserving context compression [80,81,87,88,89,134,135], continuous non-greedy constraint solving [41,42,49,50,136,137,138,139], and a computability horizon set by the undecidability of stabilization [73,74,75,76,77,144,145,146,147]. Related recent work unifies these threads via categorical deep learning [101] and sheaf neural networks [110,111,112,113].

0. Claim map: how the theory and the solver line fit

This paper is deliberately integrated: it states a lens, builds one geometric construction for it, and reports a solver line that instantiates it. Keeping the parts together is a claim about the *program*, not a licence for one part to borrow another’s credibility. Each claim therefore carries its own evidence tier.

Tier	Claims	Evidence	Does not licence
Background (standard)	quotient universal property (§2); ultrametric from refinement (§3.1); Fisher/natural-gradient flow and Lyapunov dissipation (§5.1); multiplicative gating (§6.1); undecidability horizon (§7.4)	published theorems [1,2,6,7,13,14,15,16,19,20,21,22,24,25,32,52,59,73,75,76]	novelty
New mathematics	cusps “well of uncertainty” and quadratic gradient quenching (§3.2); discovery-word lemma (§3.1); transfer criterion (§5.4)	proofs in-text [Eqs. §3.1–3.2]; convexity clause provable on Horn/submodular; parity obstruction from SOS lower bounds [166–174]	general necessary-and-sufficient characterization of when a relaxation realizes the quotient (open)
Proved (quantitative)	Chebyshev contraction and quotient-recovery rate (§5.5, Prop. 5.2, Cor. 5.1): identification in $O(\sqrt{\kappa} \log(D/\rho_0))$	gradient-polynomial minimax proof [178]; requires a filtered method [175,176,177]	the quotient identification itself — conditional on clauses A and B of §5.4
Proved (conditional)	Thm 5.2 conditional quotient realization: basin map is a well-defined surjection; recovery in $O(\frac{1}{\mu} \log(D/\rho_0))$ flow time (or $O(\sqrt{\kappa} \log)$ filtered)	proof in-text; attractor structure via Conley [179]	hypothesis (A) for a concrete class — open; submodular gives only the coarsest quotient [180]
Worked instances	1-D double well: A1 holds, proved; 2-D Gaussian well: A1 fails for $A > \sigma^2/2$ by a saddle-node birth of a spurious minimum (§5.7)	explicit critical-point and Hessian classification	a <i>rich</i> -quotient class with no surplus index-0 components

Tier	Claims	Evidence	Does not licence
Operational weakening	A1' navigational accessibility (§5.8): the quotient is realized by branch-jump navigation over the Morse decomposition; BAHA operationalizes it	fracture = fold, Lambert-W = fold continuations [61,154,155,162]	A1' proved for a class; false-positive control of the detector
SOS-certified A1	Lemma 5.1 (§5.7.3): a Positivstellensatz certificate of no unlabeled index-0 critical points proves A1 exactly on the labeled wells; the fold $\frac{x^3}{3} - \mu x$ is the birth mechanism	certificate searched in the Lasserre/SOS hierarchy [55,56,182,183,184]	a <i>rich</i> -quotient class whose certificate exists at bounded degree
Externally verified self-evaluation	solver correctness: clause-checked assignments; 10/10 colouring; client-verified hospital schedule; planted $n = 400, \alpha = 4.0$ 5/5; BAHA's solved instances; FUTCache exact novelty set $U_\varepsilon(H)$	independent checkers [154,155,156,157,158]; brute-force oracle equivalence, 134 C tests, exhaustive optimum for $n \leq 16$ [159]	comparative SOTA; generalization; that the geometry <i>caused</i> the results
Self-measured	BAHA effect sizes (+18%, attribution); SUTRA-vs-CP-SAT 9/10; throughput/scaling; 99-100% band; FUTCache 1M-trace suppression and CDCL geometry prediction	self-run harness [160]	anything beyond those specific runs
Conditionally proved (formalized)	Riemann-lock algebra (§5.3): downstream implication machine-checked in Lean 4, no sorry/axiom/admit [164]	Lean proof; the hard inputs are named hypotheses , not discharged	that the hypotheses hold
Conjecture	prime weighting (§5.2)	consistency arguments	reliability

Tier	Claims	Evidence	Does not licence
Proposal (no experiments)	evidence compression (§7.2)	—	empirical status

The integration thesis is narrow and, in principle, testable: the same design criterion — *forget when distinctions cease to matter; decide when alternatives cease to differ* — organizes both the abstraction theory and the solver’s state/commitment policy. The solver line is a **reference implementation** of that policy, and its correctness results are externally verified. It is not evidence for the quotient universal property (which is standard), and the universal property is not evidence that the solver works. Connecting them is the open programme (transfer theorem, §D.3.1; mechanism ablation, gap G7).

1. Introduction: premature commitment

Complexity is usually framed as scaling in time and space. The operative failure mode is different: **premature commitment**. A DPLL/CDCL solver assigns a variable on local evidence (activity counters, phase heuristics) and irrevocably bisects the space; near the random 3-SAT clustering threshold $\alpha \approx 4.267$ these miscommitments multiply exponentially and the search fragments [3,41,42]; the clustering transition separating the easy and hard phases was located by entropy computations [43] and characterized via Gibbs states [44]. The opposite pathology is indiscriminate retention: keeping the full history $H_t = (a_0, \dots, a_t)$ exhausts memory, while recency-based forgetting (LRU, sliding attention windows [86], greedy beam pruning) discards distinctions that still carry predictive power, producing catastrophic forgetting [7,8,80,84].

Thesis. *Computation is progressive commitment under generated geometry.* Each step is dual: **forget** distinctions that can no longer affect any admissible future (the future-indistinguishability kernel), and **decide** by pruning futures incompatible with accumulated constraints. The geometry of the state space is not an ambient container; it is generated by distinguishability. Both halves are classical: forgetting is quotienting by the right congruence [1,2] / causal equivalence [6,7] / bisimulation [15,16] / the best abstraction of a Galois connection [17,18]; deciding is the restriction of admissible continuations. What is assembled here is the unification across the discrete, continuous, and memory regimes.

2. The universal factorization theorem

Setup. Let $\mathcal{H}$ be histories, $\mathcal{W}$ continuations, $\mathcal{O}$ outcomes, and $F : \mathcal{H} \times \mathcal{W} \rightarrow \mathcal{O}$ the evaluation map [13,21,22]; this is the predictive-state / POMDP view of state [95,96,97].

Definition 2.1 (future equivalence). $h_1 \sim_{\mathcal{W}} h_2$ iff $\forall w \in \mathcal{W}, F(h_1, w) = F(h_2, w)$. Write $\mathcal{M}^* = \mathcal{H} / \sim_{\mathcal{W}}$ with projection Q .

Definition 2.2 (future-sufficient representation). A pair (M, R) with $R : \mathcal{H} \rightarrow M$ is future-sufficient if some $\Phi : M \times \mathcal{W} \rightarrow \mathcal{O}$ satisfies $\Phi(R(h), w) = F(h, w)$.

Theorem 2.1 (universal factorization). (1) $(\mathcal{M}^*, Q)$ with $\tilde{F}([h], w) := F(h, w)$ is future-sufficient. (2) For every surjective future-sufficient (M, R) there is a unique $\phi : M \rightarrow \mathcal{M}^*$ with $Q = \phi \circ R$.

Proof. (1) Well-definedness is exactly the substitution property of an equivalence relation. (2) Define $\phi(m) = Q(h)$ for any $h \in R^{-1}(m)$; future-sufficiency forces $F(h_1, \cdot) = F(h_2, \cdot)$ on the fibre, so ϕ is well-defined; commutativity and uniqueness are immediate. ■

This is not a new theorem: it is the universal property of the **coequalizer** of the kernel pair in **Set** [19,20], and simultaneously the **minimal realization** theorem for deterministic systems [21,22]; the same universal property organizes categorical deep learning [101,102,103] and geometric deep learning [151]. Specialized:

Setting	Equivalence	Quotient
DFA / regular language	$x \equiv_L y$ iff $\forall z, xz \in L \Leftrightarrow yz \in L$	minimal DFA [1,2,3,4,5]; cascade decomposition [23]
Stochastic process	same conditional future distribution	causal states / ϵ -machine [6,7]; PSRs [95]
Labeled transition system	bisimulation	bisimulation quotient [15,16]
MDP (exact / approximate)	same value & dynamics; quantitative bisimulation metrics	exact aggregation [9,10]; bisimulation metrics [11]; near-optimal guarantees [12]; homomorphisms [98,99]
Abstract interpretation	Galois connection (α, γ)	best abstraction α [17,18]

Corollaries. Minimal sufficient representations are unique up to unique isomorphism; redundant distinctions are $\ker \phi$; and refining is safe while coarsening is not quotiented correctly [1,2,17,18]; these quotients are computed by partition refinement [4,100].

Monotone coarsening. If $\mathcal{W}_{t+1} \subseteq \mathcal{W}_t$ then $h_1 \sim_{\mathcal{W}_t} h_2 \Rightarrow h_1 \sim_{\mathcal{W}_{t+1}} h_2$. Hence commitment shrinks $\mathcal{W}$, coarsens $\sim_{\mathcal{W}}$, and *enlarges* equivalence classes, licensing erasure. Deciding without coarsening is exactly the premature-commitment error [3,41,42].

Forget when distinctions cease to matter; decide when alternatives cease to differ.

3. Generated geometry

3.1 Discrete regime: projective limits generate an ultrametric

Let $\mathcal{P}_0 < \mathcal{P}_1 < \dots$ be a refining sequence of partitions of an execution space. For a trace L let $D_j(L)$ be its **discovery word** — cells of $\mathcal{P}_j$ in order of first encounter, re-entries deleted. Refinement induces bonding maps $q_j : X_{j+1} \rightarrow X_j$, which are compatible: $q_j(D_{j+1}(L)) = D_j(L)$. This is the standard **trace / partial-order** projection: discovery words are normal forms of interleavings under a Mazurkiewicz trace equivalence [28]; partial-order reduction computes representatives of these classes [29], and event structures supply the underlying concurrent semantics [30]. The inverse system has projective limit $X_\infty = \varprojlim X_j$, compact Hausdorff and non-empty by Tychonoff [20,26].

Separation index $k(x, y) = \inf\{j : x_j \neq y_j\}$ and generated metric $d(x, y) = 2^{-k(x, y)}$ give the strong triangle inequality $d(x, z) \leq \max\{d(x, y), d(y, z)\}$; hence (X_∞, d) is a complete, totally disconnected **ultrametric** space (all balls clopen).

Tree identification. Building the prefix tree $\mathcal{T}$ with vertices $\bigcup_j X_j$, the Gromov product equals lowest-common-ancestor depth, $(u \mid v)_o = \text{depth}(u \wedge v)$, and the visual metric on the Gromov boundary is $d_{\partial\mathcal{T}}(\xi, \eta) = 2^{-k(\xi, \eta)}$ [24,25]. So $(X_\infty, d) \cong \partial\mathcal{T}$ isometrically: **distance is accumulated failure of indistinguishability**. This is the same construction as the ultrametric of a profinite group [26] and of a hierarchical clustering dendrogram [27]; the ultrametric geometry of replica-symmetry-broken spin glasses [92,93] is the physical instance, and ultrametric embedding is a standard ML primitive [108,109].

3.2 Continuous regime: the uncertainty simplex and its completion

Relax $x_i \in \{0, 1\}$ to $x_i \in [0, 1]$, with $x_i = \frac{1}{2}$ maximal uncertainty and $x_i \in \{0, 1\}$ certainty. The natural geometry on this simplex is not Euclidean but the **Fisher-Rao** metric

$$g_{ii}(x) = \frac{1}{x_i(1-x_i)}, \quad g^{ii}(x) = x_i(1-x_i),$$

the unique (Chentsov) invariant metric of the Bernoulli family [31,32,33]. The induced gradient flow is **natural gradient / entropic mirror descent**,

$$\dot{x}_i = -g^{ii}(x_i) \partial_{x_i} F = -x_i(1-x_i) \partial_{x_i} F,$$

whose inverse metric vanishes at the certainty vertices [32,34,35]. The map $z_i \mapsto x_i$ with $z_i = 0 \leftrightarrow x_i = \frac{1}{2}$, $|z_i| = 1 \leftrightarrow x_i \in \{0, 1\}$ embeds the simplex in the disk $\mathbb{D}$.

Consistency erratum (original §3.4-3.5 vs §5.2). The original paper uses two different inverse metrics: $g^{z\bar{z}} = \frac{1}{4}|z|^2(1-|z|^2)^2$ (vanishing at the *centre*) and $g^{ii} = x_i(1-x_i)$ (vanishing at the *vertices*). These are not the same geometry. We record both roles: the Fisher metric gives the standard, well-posed natural-gradient dynamics of §5; the cusp conformal metric below gives a design-time “well of uncertainty” and should not be conflated with it.

The **cusp completion** with conformal factor $\Omega(z) = 2/(|z|(1-|z|^2))$, i.e. $ds^2 = 4|dz|^2/(|z|^2(1-|z|^2)^2)$, is a complete hyperbolic-type metric on $\mathbb{D} \setminus \{0\}$ with a cusp at the origin [25,36,104]; hyperbolic embeddings are now standard for hierarchical data [105]. The radial length $\int_\epsilon^R \frac{2 dt}{t(1-t^2)} = \ln(\frac{t^2}{1-t^2})|_\epsilon^R \rightarrow \infty$ as $\epsilon \rightarrow 0^+$: the origin (maximal uncertainty) is at **infinite geodesic distance** from every interior point. Under this metric the Riemannian gradient quenches quadratically,

$$\|\nabla_g E\| = g^{z\bar{z}} \|\nabla E\| = \frac{|z|^2(1-|z|^2)^2}{4} \|\nabla E\| = O(|z|^2),$$

so isolated or weak constraints exert vanishing drive at maximal uncertainty; only collective constraint pressure of order $|z|^{-2}$ can lift a variable out of the well. This is a *designed* barrier, not a theorem about SAT; the standard, physically grounded realization of the same “slow near the boundary” effect is natural-gradient/mirror descent [32,34,35]; piecewise-linear relaxations admit a tropical-geometry reading [106], and distances between such generated geometries are measured by optimal transport [107].

4. Compatibility and cohomological obstruction

Let $V = \{x_1, \dots, x_n\}$ and constraints $\mathcal{C}$ over variable supports. The **presheaf of local solutions** $\mathcal{F}$ assigns to $U \subseteq V$ the partial assignments satisfying all constraints fully inside U , with restriction maps $\rho_U^W(\sigma) = \sigma|_U$. Global solutions are global sections $\Gamma(V, \mathcal{F}) = \varprojlim_U \mathcal{F}(U)$; a solution exists iff $\Gamma \neq \emptyset$. This is the standard sheaf formulation of constraint networks [37,38,39,40]; sheaf neural networks generalize graph convolution via the sheaf Laplacian [111]; neural sheaf diffusion shows that the learned sheaf structure controls heterophily and oversmoothing [110]; connection-Laplacian variants learn the restriction maps [112]; and [113] surveys the area. Sheaf cohomology can itself encode counting hardness [114].

For a cover $\mathcal{U} = \{U_i\}$ and a coefficient module $\mathcal{A}$ (e.g. $\mathbb{Z}_2$), pairwise compatibility is the 1-cocycle condition $\delta^0 \sigma = 0$, with $(\delta^0 \sigma)_{ij} = \rho_{U_{ij}}^{U_j} \sigma_j - \rho_{U_{ij}}^{U_i} \sigma_i$. Failure of local-to-global gluing is measured by the Čech class $[\omega] \in \check{H}^1(\mathcal{U}, \mathcal{A}) = \ker \delta^1 / \text{im } \delta^0$ [37,38,39]. The same formalism detects **contextuality** obstructions in physical systems [37].

Discipline. Cohomology is a language for *obstructions to section extension*, not a complexity lower bound. Satisfiable instances can be exponentially hard (cryptographic/planted problems), and unsatisfiable instances with $[\omega] \neq 0$ can be easy: 2-SAT is linear-time solvable [Aspvall-Plass-Tarjan], Horn-SAT and GF(2) linear systems are polynomial [40, Schaefer]. Hence

The research question is not “NP-hard $\equiv \check{H}^1 \neq 0$ ” but “what dynamics bypass a non-zero cocycle?”

5. Dynamical relaxation and spectral competition

5.1 Free energy and Lyapunov stability

Let $x \in (0, 1)^n$ and

$$F[x] = \lambda E_{\text{kin}}[x] + E_{\text{pot}}[x] - \frac{1}{\beta} S[x],$$

with graph-Dirichlet kinetic term $E_{\text{kin}} = \frac{1}{2} \text{Tr}(x^\top L x)$ ($L = D - A$ the constraint-graph Laplacian [56,57,58]), a log-barrier potential

$$E_{\text{pot}} = - \sum_{c=1}^M W(p_c) \ln(\epsilon_0 + 1 - \prod_{i \in c} L_i(x_i)),$$

and Bernoulli entropy $S[x] = - \sum_i (x_i \ln x_i + (1 - x_i) \ln(1 - x_i))$. The flow $\dot{x}_i = -x_i(1 - x_i) \partial_{x_i} F$ is the natural-gradient/mirror flow of §3.2 [32,34,35,47]; nonconvex convergence guarantees for it are established in [118]. Loss landscapes of this kind share glassy statistical-mechanics structure with disordered systems [115,116], whose stiffness controls generalization [117].

Theorem 5.1 (monotone dissipation). F is a strict Lyapunov function: $\dot{F} = - \sum_i x_i(1 - x_i)(\partial_{x_i} F)^2 \leq 0$, with equality iff $\nabla F = 0$; entropy forces $-\frac{1}{\beta} \partial_{x_i} S \rightarrow \mp \infty$ at $x_i \rightarrow 0^+/1^-$, so trajectories are confined to the open cube and converge to the largest invariant set inside $\{\dot{F} = 0\}$ by **LaSalle’s invariance principle** [47,59,60]. Landauer’s bound fixes the erasure cost [119], reversible computing shows that bound is approachable [120], and the free-energy principle [121] supplies a competing variational formulation; predictive-state entropy carries a known thermodynamic cost [122]. The log-barrier is precisely Nesterov-Nemirovskii self-concordant barrier theory; $\epsilon_0 > 0$ is a standard smoothing [47,48]. ■

5.2 Prime weighting: anti-resonance by linear independence

Assign clause c weight $W(p_c) = (1 + \ln p_c)^{-1}$ for the c -th prime p_c .

Proposition 5.1. $\{\ln p_i\}$ is $\mathbb{Q}$ -linearly independent.

Proof. $\sum q_i \ln p_i = 0$ clears to $\prod p_i^{q_i} = 1$; unique factorization forces all $q_i = 0$. ■ [72,71]

This is exactly the hypothesis of the **Kronecker-Weyl equidistribution theorem**: independent logs give dense, non-resonant joint phases, so the weighted force field cannot phase-lock into accidental harmonic cancellation [71,72]. Equivalently, in log-domain the clause sum is a **product of experts** [52] and a deterministic instance of **multiplicative weights** [54,55].

Correction. The original claims $\sum_{c \leq K} W(p_c) \sim K$. By the prime number theorem, $W(p_c) \asymp 1/\ln c$ and $\sum_{c \leq K} 1/\ln c = \Theta(K/\ln K)$, so the correct asymptotic is $M_K = \Theta(K/\ln K)$; the linear claim is off by a factor $\ln K$ [72].

5.3 Spectral gap vs. arithmetic fluctuation

Two exponents govern large-scale stability:

1. **Geometric weakening.** Algebraic connectivity (Fiedler value) decays as $\lambda_2(G_K) \asymp K^{-\gamma}$; Fiedler's algebraic connectivity defines the quantity [56], Cheeger-type inequalities relate the gap to graph expansion [57], and mixing-time bounds connect spectral separation to diffusion rates [58].
2. **Prime fluctuation.** By the explicit formula, $\pi(x) = \text{Li}(x) + \mathcal{O}(x^\sigma \log x)$ with $\sigma = \sup\{\text{Re } \rho : \zeta(\rho) = 0\}$; the relative weight variance scales as $\Phi(K) \asymp K^{\sigma-1}$ [67,68].

Stability criterion (heuristic). Linearizing the slowest graph mode, $\delta \dot{x}_2 = -\lambda_2(G_K) \delta x_2 + \Xi(K)$; the perturbation is dominated when $\Phi/\lambda_2 \asymp K^{\sigma-1+\gamma} \rightarrow 0$, i.e.

$$1 - \sigma > \gamma.$$

Unconditional anchor. Bombieri-Vinogradov gives averaged equidistribution of primes in progressions to level $\theta = \frac{1}{2}$, i.e. an averaged $\Phi(K) \asymp K^{-1/2}$ without RH [64,65,66]; prime fluctuations are themselves random-matrix distributed at leading order [123,124,125], with unconditional regularity from short-interval and prime-gap results [126,127]. Two competing spectral interpretations of the zeros are Connes' trace-formula program [69] and the Berry-Keating quantum-chaos heuristic [70].

The Riemann lock. If a constructible constraint family achieves the critical barrier $\gamma \rightarrow \frac{1}{2}$, then $1 - \sigma > \frac{1}{2}$ forces $\sigma < \frac{1}{2}$; since zeros on the critical line exist (Hardy) and the functional equation pairs $\rho \leftrightarrow 1 - \bar{\rho}$, stability then requires $\sigma = \frac{1}{2}$ — the Riemann Hypothesis. The Bost-Connes system provides an arithmetic model of exactly this kind of phase transition [94]. This is a **conditional equivalence, not a proof**, and it remains open whether such a family exists [67,68]. The downstream implication is now **machine-checked**: `ConditionalAsymptoticLock.lean` proves, with no sorry, axiom, or admit, that *under* (i) the linear-coupling threshold equality and (ii) the existence of a critical graph family, asymptotic stability on that family is equivalent to $\sigma \leq \frac{1}{2}$ [164]. Its status is therefore **conditionally proved**, not conjectural: the exponent algebra is verified, while the analytic/spectral hypotheses — and the existence of the critical family — remain the open inputs [67,68].

5.4 Transfer criterion: when a relaxation realizes the quotient

Criterion. A continuous relaxation can realize a behavioral quotient when the discrete problem admits an exact convexity preserved by the relaxation, the observables separate quotient classes, and the resulting basins are uniformly conditioned; parity-type affine structure provides a canonical obstruction.

The three clauses are the hypotheses needed for the three obligations of the transfer theorem (Appendix D.3.1):

1. **Convexity preserved (correctness).** The energy must admit an exact convex or discrete-convex extension whose minimizers are integral: median/CAT(0) closure (majority polymorphism, 2-SAT/bijunctive) [168,171], lattice semilinearity (Horn/dual-Horn), submodularity via the Lovász extension [169], or M/L-convexity [170]; bounded treewidth admits exact junction-tree decomposition [174]. This is the discrete version of “generated geometry” made literal — the solution set is a CAT(0) cube complex — and only here does the closure above buy exactness rather than analogy.
2. **Separation (injectivity).** The observables in $\mathcal{W}$ must distinguish every $\mathcal{W}$ -inequivalent pair; otherwise distinct quotient classes share a basin and the metric bound fails regardless of how well the flow behaves.
3. **Uniform conditioning (computability and basin refinement).** A uniform isolation radius and spectral gap on each basin, together with bounded degree/curvature, yield the computable bi-Lipschitz constants. §5.1 gives global dissipation but not this uniform gap, which is precisely what the $1 - \sigma > \gamma$ heuristic of §5.3 was reaching for.

Canonical obstruction. Affine structure (parity/XOR) is tractable but not relaxable: it is closed under no median, lattice, or submodular convexity in $[0, 1]^n$, and its LP/SOS relaxations have degree $\Omega(n)$ [172,173]. Tractability of the discrete problem is therefore necessary but not sufficient. More generally, a no-free-lunch boundary applies: if the transfer holds on a class $\mathcal{C}$ and equilibria are findable in polynomial time, then $\text{CSP}(\mathcal{C}) \in \text{P}$, so the realizable class lies inside the tractable side of Schaefer’s dichotomy [166,167] and is in general strictly smaller.

Consequence for the thesis. Geometry is not *generated* by distinguishability in general; it is *available* exactly when the discrete problem is already convex in the right sense. Where it is, a relaxation realizes the quotient; where it is not — general 3-SAT, parity at low degree — no convex relaxation can.

5.5 Quantitative recovery rate: conditioning $\Rightarrow$ computable identification

The third clause of §5.4 is quantitative, and on a quadratic basin its exact rate is a Chebyshev minimax problem — the clean object that the spectral heuristic of §5.3 was groping for.

Setup. Let a behavioral class m have representative attractor x_m , let $e_0 = x_0 - x_m$, and suppose that on its basin $F(x) = \frac{1}{2}\langle x - x_m, H(x - x_m) \rangle$ with $H = H^\top > 0$ and $\mu I \leq H \leq LI$, $\kappa = L/\mu$. Any degree- t gradient-polynomial method with fixed point x_m has error $e_t = p_t(H)e_0$ with $\deg p_t \leq t$ and $p_t(0) = 1$ (the normalization says a zero-curvature mode is left unchanged). Hence $\|e_t\| \leq \max_{\lambda \in [\mu, L]} |p_t(\lambda)| \|e_0\|$, and the optimal worst-case contraction is

$$\inf_{\deg p \leq t, p(0)=1} \max_{\lambda \in [\mu, L]} |p(\lambda)|.$$

Proposition 5.2 (Chebyshev contraction). The unique minimax polynomial is

$$p_t^*(\lambda) = \frac{T_t\left(\frac{L + \mu - 2\lambda}{L - \mu}\right)}{T_t\left(\frac{L + \mu}{L - \mu}\right)},$$

and $\max_{\lambda \in [\mu, L]} |p_t^*(\lambda)| \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^t$. Consequently

$$\|x_t - x_m\| \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^t \|x_0 - x_m\|.$$

Proof. Put $z(\lambda) = \frac{L+\mu-2\lambda}{L-\mu}$, so $[\mu, L] \rightarrow [-1, 1]$ and $\lambda = 0 \mapsto a = \frac{\kappa+1}{\kappa-1} > 1$. The problem becomes $\inf_{\deg q \leq t, q(a)=1} \max_{z \in [-1, 1]} |q(z)|$, whose unique solution is $q_t^* = T_t/T_t(a)$ (Chebyshev extremal property [178]). Since $|T_t| \leq 1$ on $[-1, 1]$, the value is $1/T_t(a)$; and $T_t(a) = \cosh(t \operatorname{arcosh} a) \geq \frac{1}{2}(a + \sqrt{a^2 - 1})^t$ with $a + \sqrt{a^2 - 1} = \frac{\sqrt{\kappa+1}}{\sqrt{\kappa-1}}$. ■

Corollary 5.1 (Chebyshev Quotient-Recovery). If representatives of distinct classes satisfy $\|x_m - x_{m'}\| \geq \rho_0$ ($m \neq m'$), the balls $B(x_m, \rho_0/2)$ are disjoint, and the class is uniquely identifiable once $\|x_t - x_m\| < \rho_0/2$. Writing $D = \|x_0 - x_m\|$, this holds after

$$t = O\left(\sqrt{\kappa} \log \frac{D}{\rho_0}\right) = O\left(\sqrt{L/\mu} \log \frac{D}{\rho_0}\right)$$

optimally filtered first-order iterations.

Scope (where this can be attacked). (i) The exact residual-polynomial argument is a theorem for a *quadratic* basin, i.e. linearized dynamics with a fixed self-adjoint Hessian; for a general nonlinear strongly convex basin the rate $O(\sqrt{\kappa} \log(1/\varepsilon))$ still holds but via accelerated methods [176], not this proof. (ii) The $\sqrt{\kappa}$ rate requires a *filtered* first-order method (Chebyshev semi-iteration or conjugate gradients [175, 177]); plain gradient descent — and the continuous flow $\dot{x} = -g^{-1} \nabla F$ of §5.1 — contracts at $1 - O(1/\kappa)$, a factor $\sqrt{\kappa}$ slower. The lemma therefore does not apply to the flow as written. (iii) It recovers the class *given* the attractor-class correspondence (clause A of §5.4); it does not establish that correspondence. It is the quantitative half of C, conditional on A and B.

conditioning κ + separation $\rho_0 \implies$ quotient identification in $O(\sqrt{\kappa} \log(D/\rho_0))$

What the spectral heuristic was reaching for. The useful object is the minimax polynomial T_t over the relaxation spectrum, not the zeta-zero exponent σ ; §5.3's $1 - \sigma > \gamma$ is a proxy for κ being uniformly bounded.

5.6 Conditional quotient realization (the transfer theorem)

The three clauses of §5.4 compose into a conditional theorem whose only non-trivial hypothesis is exactness (A).

Theorem 5.2 (conditional quotient realization). Let Φ be a finite constraint system with behavioral quotient $Q : X \rightarrow \mathcal{M}^*$, and let (K, F_Φ) be a C^2 relaxation with rounding $r : K \rightarrow X$ and dynamics $\dot{x} = -\nabla F_\Phi$ (projected to K). Assume:

- **(A) Exactness.** Every reachable attracting component A_m of the flow is associated with exactly one class m , with $r(A_m) \subseteq Q^{-1}(m)$; and every class has a reachable component with non-empty basin.
- **(B) Separation.** $\operatorname{dist}(A_m, A_{m'}) \geq \rho_0 > 0$ for $m \neq m'$.
- **(C) Conditioning.** On a neighbourhood U_m of each A_m , $\mu I \leq \nabla^2 F_\Phi \leq LI$.

Then:

1. The basin map $\pi_\Phi(x) = m$ for $x \in B_m$ is well-defined, and $\pi_\Phi : \bigsqcup_m B_m \rightarrow \mathcal{M}^*$ is a surjection.
2. For any m and $x(0) \in U_m$ with $x_m \in A_m$, $\|x(t) - x_m\| \leq e^{-\mu t} \|x(0) - x_m\|$, so the class is identified after $t = O(\frac{1}{\mu} \log(D/\rho_0))$ flow time with $D = \|x(0) - x_m\|$; with a filtered first-order method the conditioning dependence sharpens to $O(\sqrt{\kappa} \log(D/\rho_0))$ (Prop. 5.2, Cor. 5.1).

Proof. (1) If $x \in B_m \cap B_{m'}$ then $\omega(x) \subseteq A_m \cap A_{m'} = \emptyset$ by (B), so $m = m'$; surjectivity is (A). (2) On U_m , F is μ -strongly convex and $\nabla F(x_m) = 0$, so $\frac{d}{dt} \frac{1}{2} \|x - x_m\|^2 = -\langle x - x_m, \nabla F(x) - \nabla F(x_m) \rangle \leq -\mu \|x - x_m\|^2$; Grönwall gives the exponential bound [60]. Separation makes $B(x_m, \rho_0/2)$ a unique-class certificate, and $e^{-\mu t} D < \rho_0/2$ gives the time. The $\sqrt{\kappa}$ form is Prop. 5.2. ■

Attractor structure is Conley's. The “attracting components” above are exactly isolated invariant sets in a Morse decomposition, and the basin map is the associated Morse-order projection [179]; (B) is isolation plus separation.

All the content is in (A). (B) is an isolation assumption and (C) is local; (A) is the entire transfer claim. Note also that (C) is strong convexity *per basin*, not global convexity: F may be non-convex, and must be, if the quotient is to be rich (see below).

A degenerate instance, and the trap. Submodular energies are the cleanest exact case: for submodular f the Lovász extension $\hat{f}$ is convex and $\min_{[0,1]^n} \hat{f} = \min_S f(S)$, with layer-cake thresholding of a minimizer yielding discrete minimizers for almost every threshold [169]. But a convex $\hat{f}$ has a *single* convex minimizer set, hence a single attracting component. So a submodular relaxation realizes the quotient only when $|\mathcal{M}^*| = 1$ — the coarsest quotient. A rich quotient requires a non-convex F with several basins, which means (A) must be established by Morse-theoretic rather than convexity arguments [180]: non-degenerate critical points with index 0 in bijection with $\mathcal{M}^*$, and no critical point whose rounding lies outside a satisfying class. That is the honest target; the submodular case is the warm-up, not the bridge.

5.7 Two worked non-convex instances: A1 holding and failing

5.7.1 One dimension: A1 holds Let $X = \{-1, +1\}$, $\mathcal{M}^* = \{[-1], [+1]\}$, and $F(x) = \frac{1}{4}(x^2 - 1)^2$, with flow $\dot{x} = -F'(x) = x(1 - x^2)$. Then $F'(x) = x(x^2 - 1)$ vanishes only at $\{-1, 0, +1\}$, and $F''(x) = 3x^2 - 1$ gives $F''(\pm 1) = 2 > 0$, $F''(0) = -1 < 0$. Hence $\text{Crit}_0(F) = \{-1, +1\}$ with the origin an index-1 saddle whose stable manifold is $W^s(0) = \{0\}$; the basins are $W^s(-1) = (-\infty, 0)$ and $W^s(+1) = (0, \infty)$. With the rounding $r(x) = \text{sign}(x)$ the diagram $Q \circ r = \pi \circ \omega$ commutes on $\mathbb{R} \setminus \{0\}$, so $\text{Crit}_0(F) \cong \mathcal{M}^*$. Crucially, “no spurious minima” is **proved**, by solving $F' = 0$ and classifying every root through F'' — not assumed. Surplus index-0 components are the only way A1 can fail.

5.7.2 Two dimensions: a deliberately created spurious minimum (A1 fails) Let $X = \{-1, +1\}^2$ with the four singleton classes, and

$$F_{A,\sigma}(x, y) = \frac{1}{4}(x^2 - 1)^2 + \frac{1}{4}(y^2 - 1)^2 - A e^{-(x^2+y^2)/\sigma^2}.$$

The origin is critical for every A , with

$$\nabla^2 F_{A,\sigma}(0, 0) = \left(-1 + \frac{2A}{\sigma^2}\right)I.$$

So the origin is an index-0 **minimum** exactly when $A > A_c := \sigma^2/2$, and an index-2 maximum when $A < A_c$. The four corners remain non-degenerate minima whenever $A e^{-2/\sigma^2}$ is small; e.g. at $\sigma = \frac{1}{2}$, $A = 1$ the corner Hessian is $\begin{pmatrix} 1.98 & -0.02 \\ -0.02 & 1.98 \end{pmatrix} > 0$.

Therefore for $A > A_c$, $\text{Crit}_0(F_{A,\sigma}) \supseteq \{(\pm 1, \pm 1)\} \cup \{(0, 0)\}$ has at least five elements while $|\mathcal{M}^*| = 4$. No $\pi : \text{Crit}_0 \rightarrow \mathcal{M}^*$ is a bijection; the origin has a positive-measure basin and rounds to no behavioral class, so A1 fails and the relaxation does not realize the quotient.

The threshold A_c is a **saddle-node bifurcation**: the origin's Hessian eigenvalue crosses zero, and a spurious well is born there [181]. This is the generic mechanism by which a continuous relaxation acquires a spurious attractor, and hence the generic obstruction to A1.

A1 holds while $\# \text{Crit}_0(F) = |\mathcal{M}^*|$ with correct rounding; a surplus index-0 component breaks it.

The 1-D instance satisfies this; the 2-D instance violates it for $A > A_c$. The general task is thus to control the perturbation so that no surplus well is born — which is a concrete analytic obligation, not an assumption.

5.7.3 SOS-certified realization: A1 proved, not assumed The two instances above were analyzed by hand. The general statement is a *certificate* condition, which makes A1 checkable rather than assumed.

Lemma 5.1 (SOS-certified Morse realization of a finite quotient). Let $F : \mathbb{R}^n \rightarrow \mathbb{R}$ be a coercive polynomial and let $W_1, \dots, W_k$ be disjoint semialgebraic neighbourhoods labeled by the classes $m_1, \dots, m_k$ of $\mathcal{M}^*$, with $k = |\mathcal{M}^*|$. Suppose:

1. each W_i contains exactly one critical point x_i of F ;
2. $\nabla^2 F(x_i) > 0$, so x_i has Morse index 0;
3. there is a Positivstellensatz certificate that the set

$$\left\{ x \notin \bigcup_i W_i : \nabla F(x) = 0, \nabla^2 F(x) \geq 0 \right\}$$

is empty;

4. the rounding r is constant on each attracting basin and $r(W_i) \subseteq Q^{-1}(m_i)$.

Then $\text{Crit}_0(F) = \{x_1, \dots, x_k\}$, and $\pi : \text{Crit}_0(F) \rightarrow \mathcal{M}^*$, $\pi(x_i) = m_i$, is a bijection. So A1 holds.

Proof. By (1)-(2), each x_i is a non-degenerate local minimum and these are the only critical points of F inside the W_i . By (3), every critical point outside $\bigcup_i W_i$ has a negative-curvature direction and is therefore not a local minimum. Hence the local minima of F are exactly $\{x_i\}$. Clause (4) makes the class assignment well-defined on basins; with $k = |\mathcal{M}^*|$ and distinct labels, π is a bijection. ■

The certificate. Since $\nabla F = 0$ are polynomial equalities and $\nabla^2 F \geq 0$ is semialgebraic (all principal minors ≥ 0), the excluded set is semialgebraic and its emptiness can be certified by an identity

$$-1 = \sigma_0 + \sum_j \sigma_j g_j + \sum_\ell h_\ell \frac{\partial F}{\partial x_\ell} + \text{PSD terms},$$

with the σ_j sums of squares and the g_j describing the complement of $\bigcup_i W_i$ [182,183]. In practice the certificate is searched at increasing degree in the Lasserre/SOS hierarchy [55,56,184]; coercivity of F (equivalently, an added constraint $\|x\|^2 \leq R^2$) supplies the Archimedean hypothesis.

Scope. The certificate is *sufficient*, not necessary, and its existence at a fixed degree is not guaranteed: for hard instances the required degree grows, consistent with the parity obstruction [172,173]. Lemma 5.1 therefore turns A1 into a checkable obligation whose **degree** is the complexity measure — trivial in the submodular case ($k = 1$), expensive exactly where the quotient is rich and the language is hard.

Birth of surplus attractors: the fold. As a control parameter varies, the Morse structure changes through folds, with normal form [181]

$$V_\mu(x) = \frac{x^3}{3} - \mu x, \quad V'_\mu(x) = x^2 - \mu :$$

no real critical pair for $\mu < 0$, a degenerate fold at $\mu = 0$, and for $\mu > 0$ a minimum $x_+ = +\sqrt{\mu}$ and a maximum $x_- = -\sqrt{\mu}$. The birth of a (minimum, saddle) pair is exactly the event at A_c in §5.7.2. If the newborn index-0 component carries no behavioral class, then $\# \text{Crit}_0 > |\mathcal{M}^*|$, A1 fails, and the certificate of Lemma 5.1 ceases to exist. A BAHA-style fracture detector estimates this event, and branch jumping is the runtime response (§5.8).

5.8 BAHA: operational navigation of the Morse decomposition

The failure mode of §5.7 — a surplus index-0 component, or a suboptimal basin — is exactly what BAHA detects and acts on, which gives the transfer theorem an *operational* form rather than a purely existence-theoretic one.

- **Fracture = fold.** BAHA declares a fracture when the log-partition slope $\rho = |d/d\beta \log Z|$ and its variance (the heat-capacity/susceptibility proxy, since $d^2/d\beta^2 \log Z = \text{Var}(E)$) exceed thresholds [162]. Thermodynamically this is the phase transition; dynamically it is the **fold (saddle-node)** at which $\nabla^2 F$ acquires a zero eigenvalue — the same event as A_c in §5.7.2.
- **Lambert-W branches = the two continuations through the fold.** Near criticality the mean-field order parameter obeys $ue^u = \xi$, whose real branches W_0 and W_{-1} are the two stable continuations [61]; Proposition 1 of [155] asserts they enumerate all thermodynamically stable continuations to leading order in $\beta - \beta_c$ (mean-field scope). Lambert-W is thus the normal form of the A1 transition.
- **Branch jumping = Morse navigation.** Instead of requiring a single flow to reach every index-0 component (A1), BAHA detects when its current basin is suboptimal and samples another component [162].

This motivates a weaker, more realistic accessibility condition in place of A1:

A1' (navigational accessibility). Every behavioral class is reachable from some admissible initial state by a finite sequence of gradient descents interleaved with fracture-triggered branch jumps.

If A1' holds together with the separation and conditioning (B), (C), then Theorem 5.2 still yields a well-defined class map on the *reachable* Morse components, and the quotient is realized by the navigation policy rather than by a single flow. This is the honest, implementable form of the transfer: BAHA is the search over the Morse decomposition that (A) assumes.

Empirical status, stated honestly. BAHA's own case studies report that fractures are frequent but branch jumps are selective (599 fractures / 4 jumps on VRP; 499 / 8 on network design) [162]: the detector has many false positives and the policy must filter them. That is the operational analogue of the spurious-minimum problem — separating *actionable* transitions from nuisance ones. The reported ablation gains (fracture detection alone $\approx +18\%$ over SA, [155]) are self-measured, not replicated.

fracture $\equiv$ fold, branches $\equiv$ continuations, jumping $\equiv$ Morse navigation, A1 $\Rightarrow$ A1'

6. Multiplicative gating and thermodynamic transitions

6.1 Direction preservation vs. additive cancellation

Additive penalties $\mathcal{L}_{\text{add}} = \mathcal{L}_{\text{data}} + \mu \mathcal{L}_{\text{constr}}$ produce conflicting gradients that cancel ($\langle \nabla \mathcal{L}_{\text{data}}, \nabla \mathcal{L}_{\text{constr}} \rangle < 0$), the multi-task failure mode addressed by gradient surgery [53] and surveyed in constrained-optimization learning [128]. Multiplicative gating instead sets $\mathcal{L}_{\text{mult}} = \mathcal{L}_{\text{data}} \cdot C(\nu)$, so

$$\nabla_{\theta} \mathcal{L}_{\text{mult}} = C(\nu) \nabla_{\theta} \mathcal{L}_{\text{data}} + \mathcal{L}_{\text{data}} \partial_{\nu} C \nabla_{\theta} \nu.$$

Proposition 6.1. On the subspace $\nabla_{\theta} \nu = 0$ the direction is invariant, $\nabla_{\theta} \mathcal{L}_{\text{mult}} \propto \nabla_{\theta} \mathcal{L}_{\text{data}}$; off it the data gradient is rescaled, never cancelled. This is the **product-of-experts** combination rule [52,129] in log space, i.e. an adaptive volume control rather than a competing steering wheel.

Euler gate. $G(\nu) = \prod_{p \in \mathcal{P}_{\text{gate}}} (1 + p^{\tau \nu})$ ($G(0)$ normalizable to 1) grows at rate $\sum \tau \ln p$; combined with a self-concordant exponential barrier $B(\nu) = e^{\kappa \nu}$, take $C(\nu) = \max(G(\nu), B(\nu))$. Small primes penalize macro-violations, large primes micro-violations; the barrier expels infeasible configurations without rotating the task direction [47,48,54,55].

6.2 Thermodynamic fracture and the Lambert-W jump

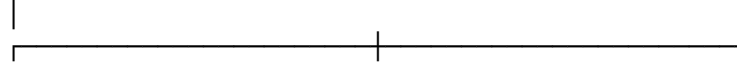
As β increases, the free-energy landscape of spin-glass-like systems fractures; the clustering transition of random k -SAT is the canonical example, and survey propagation is the standard algorithm that jumps the fracture rather than descending into it [41,42,43,44,92,93]. Writing $\Phi(\beta) = \ln Z(\beta)$, the moments are $\langle E \rangle_{\beta} = -\partial_{\beta} \ln Z$ and $\text{Var}_{\beta}(E) = \partial_{\beta}^2 \ln Z$. Near a fold $\beta - \beta_c = u$ with $ue^u = \xi$, the escape jump obeys the fold normal form of catastrophe theory [62,63] and is solved by the principal Lambert-W branch, $\Delta\beta = W_0(\xi)$ [61, Section 6.3]. Using $k! \approx (k/e)^k$ to set the Laplace horizon $k_{\text{max}} \approx T/e$ and $W_0(y) = \ln y - \ln \ln y + o(1)$ gives

$$\Delta\beta \approx \frac{T}{e} - \ln \frac{T}{e}.$$

This Laplace-horizon derivation is informal, but the *method* it motivates is not merely asserted. BAHA (Branch-Aware Exact Basin Hopping) detects such fractures and jumps branches, and has been benchmarked: on a 26-domain suite it solves 22/26 (84%); fracture detection alone improves simulated annealing by $\approx 18\%$; branch scoring accounts for roughly half the total gain; and on smooth (fracture-free) landscapes it degrades to SA within 5% overhead, with ablations reported over 200 instances at 95% confidence [154,155]. The repository also documents its own failures (XOR-SAT, smooth landscapes, an admitted ChaCha20 cryptanalysis attempt), and its current table reports 22/26 rather than the 23/26 on the hosted article [162]. The claim is therefore **empirically supported**, not proved: branch completeness is a proposition conditional on mean-field spin-glass structure [155], and the Fisher-information step above remains informal. Experiments to date compare against simulated annealing, not against state-of-the-art CDCL/survey-propagation solvers.

7. Instantiations

$M^* = H / \sim_W$ (Thm 2.1, = Myhill–Nerode / causal states)



▼ ▼ ▼
 [DISCRETE] [CONTINUOUS] [MEMORY]
 FUTCache / KV NitroSAT / SOS Evidence compression
 ε -quotient of Fisher + barrier quotient by answer-prompt states + prime gates invariance

7.1 KV cache: ε -quotienting of prompt states

History = prefix tokens; continuations = future queries/completions; outcome = completion distribution (Transformer attention is the substrate [91]). Exact caching uses the identity relation $\mathbf{x}_1 = \mathbf{x}_2$; the quotient principle replaces it with

$$\sup_{w \in \mathcal{W}} D_{\text{TV}}(P(\cdot | \mathbf{x}_1, w), P(\cdot | \mathbf{x}_2, w)) < \varepsilon,$$

which is **rate-distortion / information-bottleneck** compression of the predictive state [80,81,82] and the approximate form of causal-state minimality [6,7]. Practical systems instantiate coarse versions of this quotient: paged/block KV memory [83], heavy-hitter eviction [84], attention-sink streaming [85], sliding-window attention [86], KV quantization [130], and persistence-based eviction [131]; the broader generation-efficiency stack is surveyed in [132,133]. The quotient viewpoint predicts the right comparison: evict by *downstream equivalence class*, not by recency or norm. What a transformer can compute at all is itself a formal-language question [142,143]; recent long-context systems make the state itself a learned compression [150].

A concrete realization: FUTCache. FUTCache implements exactly this quotient for *novelty* decisions rather than token caches. The state is the union $U_\varepsilon(H) = \bigcup_{y \in H} \bar{B}(y, \varepsilon)$; two histories with the same union are future-equivalent for every later novelty query, so the raw stream can be forgotten [159]. The stored representative set is ε -separated, so its size is bounded by the packing number $P(K, \varepsilon)$ rather than by stream length. Exactness of the interval-union and box-union engines is checked against brute-force oracles (134 C tests), and the submodular selection path is checked against exhaustive optima for $n \leq 16$ — verification, not assertion [159]. Empirical receipts are self-measured: on 1,000,000 real Alibaba Cloud traces, the net suppressed 72.34% of query work at $\varepsilon = 0.55$ and grew sublinearly (65.95 MB at $N = 10^6$) [160]; a separate CDCL study reports metric clause geometry predicting exact backjump depth at 20.0% vs a 14.6% matched null ($p = 0.007$), rising to 41.3% when combined with LBD [160].

The instantiation also carries a **negative result** that sharpens the theory: choosing ε by geometric MDL is *not* semantic safety. On a labeled semantic-cache benchmark the MDL-optimal radius ($\varepsilon = 0.70$) gave F1 0.558 with 11 cross-intent merges, while the supervised optimum was $\varepsilon = 0.55$ (F1 0.743) and the zero-false-positive point was $\varepsilon = 0.45$ [161]. Geometric compressibility and semantic substitutability are distinct objectives: the quotient must be taken with respect to the *task* continuation set $\mathcal{W}$, not the cheapest description. This is the design principle in its sharpest, falsifiable form.

7.2 Answer-preserving context compression

History = retrieved documents E ; continuations = queries Q ; quotient $E_1 \sim_{\mathcal{W}} E_2 \iff \forall q \in Q, \text{Ans}(q, E_1) = \text{Ans}(q, E_2)$; optimal $E^* = \arg \min_{E' \subseteq E} |E'|$ with $Q(E') = Q(E)$. This is the exact quotient projection of documentary history modulo answer-invariance, and it specializes both the information bottleneck [80,81] and causal-state minimality [6,7]. Current prompt compressors — LLMingua/LLMingua-2 [87,88], gist tokens [89], retrieval pipelines [90,135], and the survey [134] — optimize surrogate salience or task loss; the quotient criterion makes the invariance target explicit.

7.3 Continuous non-greedy satisfaction (NitroSAT)

Map Boolean variables to the uncertainty simplex/disc (§3.2). Variables with low distinguishability pressure stay at $z_i = 0$ ($x_i = \frac{1}{2}$), where the cusp metric quenches $\nabla_g F = \mathcal{O}(|z|^2)$; the flow refuses to split until global constraint pressure lifts the variable toward certainty. This

belongs to the continuous/learned-relaxation family of SAT solvers: Hopfield energy formulations [45], Fourier/algebraic continuous solvers [139], the Goemans-Williamson SDP approximation [49], the Lasserre hierarchy [50] and sum-of-squares relaxations [51] (applied to MAX-SAT in [138]), and neural message-passing solvers [136,137]. The distinctive claim is not convexification but *when* commitment is licensed: the geometry of §3.2 plus the certificate of §4. The clustering literature [41,42,43,44] supplies the correctness target — a solver should bypass the 1RSB clustered phase the way survey propagation does [140,141]. An implementation ($\approx 2,500$ lines of LuaJIT) reports 99–100% clause satisfaction on hard structured instances, linear scaling to 200k variables at 1,200–1,600 clauses/s, and completion on an 80M-clause scheduling instance and a 4M-variable grid-coloring instance [156]. These are single-machine, self-run benchmarks — not independent replication — but the promoted runs are externally verified: the returned assignment is checked clause-by-clause against the original instance, so they are **externally verified self-evaluation** [156,157]. No experiment isolates the cusp metric, the quotient policy, or the prime gates as the *cause* of these results: the geometry is motivation for the implementation, not a tested mechanism (gap G7, §7.5).

7.4 The computability horizon

Let L run on an undecidable configuration graph. Each finite $D_j(L)$ is computable, but the coherent profile $x^* \in X_\infty = \varprojlim X_j$ is a limit object.

Theorem 7.1. No general algorithm decides whether a finite prefix $D_j(L)$ has permanently stabilized.

Proof. Encode a Turing machine T so a marked cell C_{halt} is ever visited iff T halts; a uniform stabilization test would decide the halting problem, which Turing proved undecidable [76]. ■ [73,74,75]

The quotient $\mathcal{M}^*$ exists as a compact mathematical ceiling but is not constructible by a finite machine; computation proceeds inside finite approximations $M_0 \leftarrow M_1 \leftarrow \dots \leftarrow \mathcal{M}^*$. This is the standard phenomenon of **non-computable limits of computable sequences** (Specker sequences; Pour-El-Richards; Weihrauch) [73,77,78], of **identification in the limit** [74,75] — with constructive query-learning as the positive counterpart [5] — of the incompressibility barrier of Kolmogorov complexity [79], and of learnability undecidability [147]. It sits between the super-Turing power of analog/neural models [144,145,148] and the computability limits of realistic quantized networks [146]; see [149] for the statistical backdrop. Every sound algorithm must emit finite witnesses at finite resolution.

7.5 Empirical status, under an explicit evidence standard

Two orthogonal axes govern an empirical claim. **Provenance** determines independence; **verification** determines validity. They must not be conflated:

Level	Criterion	Strength
Self-reported	solver’s own success flag	weak
Externally verified	solver emits a witness; a separate checker validates it against the original instance	valid evidence of correctness
Independently replicated	separate implementation or researcher reproduces the result	strongest

A satisfying assignment for a SAT instance, checked clause-by-clause by a verifier that reads the original CNF, is a proof that the instance is satisfiable and that the solver found a witness — it does not require a competing solver to certify the same assignment. The same standard applies to ablations and metrics: an independent checker can validate outputs even when the harness and the solver share a project. What external verification cannot establish by itself is comparative SOTA performance, freedom from benchmark-selection bias, or generalization beyond the tested distribution.

Correctness claims (verification axis). These are valid evidence that the instances are satisfiable and that the solver found a witness. They say nothing about competitors.

System	Claim	Separate verifier	Status
NitroSAT (§7.3)	promoted runs return satisfying assignments	clause-by-clause check against the original CNF [157]	valid
SUTRA / nitro	10/10 colouring & list-colouring suite	assignments independently checked against original CNFs [158]	valid
SUTRA / nitro	hospital scheduling, 504 vars / 39,842 clauses, SAT ≈ 0.15 s	client-side independent verifier [158]	valid
SUTRA / nitro	planted 3-CNF, $n = 400$, $\alpha = 4.0$, 5/5 closures	independent witness recount [158]	valid
BAHA (§6.2)	22/26 solved solutions	witness/solution checks on the reported ones [154,155,162]	valid for the solved instances
FUTCACHE (§7.1)	exact novelty set $U_\varepsilon(H)$	brute-force oracle equivalence (134 C tests); exhaustive optimum for $n \leq 16$ [159]	valid

Comparative and aggregate claims (provenance axis; verification cannot validate these). These must be labelled self-measured, not externally verified.

Claim	Status
BAHA $\approx +18\%$ over simulated annealing; branch scoring $\approx$ half the gain; within 5% on smooth landscapes	self-measured effect sizes in a self-run harness; not verified, not replicated [155]
SUTRA vs CP-SAT 9/10 within 10 s	self-run comparison on a small, unregistered suite; the checked assignments prove SUTRA's 10, not the fairness of the CP-SAT run [158]
NitroSAT throughput (1,200–1,600 clauses/s; 200k vars; 80M-clause; 4M-var completion)	self-run timing; completion is valid only where those assignments were clause-checked, which must be stated per instance [156]

Claim	Status
99-100% clause-satisfaction band	partial satisfaction is not a correctness claim; only fully checked assignments certify SAT [156]
26-domain and 10-instance generalization	suites are self-selected; not evidence beyond those instances [154,155,158]
FUTCACHE 72.34% suppression on 1M Alibaba traces; 65.95 MB at $N = 10^6$	self-measured on a public dataset; not replicated [160]
FUTCACHE CDCL backjump prediction 20.0% vs 14.6% null ($p = 0.007$); 41.3% with LBD	self-measured with a null model; not replicated [160]
FUTCACHE MDL radius is not semantic safety (F1 0.558 vs supervised 0.743)	self-measured negative result , preserved as a regression test [161]

What verification cannot establish (gaps). G1 independent replication: absent, all sources are one project. G2 comparative baselines: no CDCL (CaDiCaL/Kissat), no Survey Propagation, no modern MaxSAT (RC2), no H2O/StreamingLLM/LLMLingua. G3 benchmark-selection bias: uncontrolled; suites and “promoted runs” are not pre-registered. G4 generalization: no held-out distribution. G5 UNSAT side: no DRAT/LRAT certificates; all validity evidence is SAT-witness-only. G6 verification coverage: the fraction of runs checked, checker identity, and whether checker code is separate from solver code are unstated. G7 mechanism ablation: no experiment isolates the cusp metric (§3.2) or prime gates (§5.2) as causal. G8 one instantiation (§7.2) has no experiments; FUTCACHE is implemented and oracle-verified. G9 theory-empirics mismatch: the empirical work does not test the Laplace-horizon derivation, prime weighting, or the Riemann lock. G10 sample sizes: headline aggregates rest on 10, 5, 1, and 26 instances, one machine, one implementation.

7.6 Code and artifacts

All implementations are public and inspectable; the verification recipes live in the repositories.

Artifact	What it is	Verification present	Ref
FUTCACHE	C11 metric visited-set; exact interval-union and L_∞ box-union engines; CRDT; Python bindings	brute-force oracle equivalence (134 C tests); exhaustive optimum for $n \leq 16$; MDL negative result preserved as a regression test	[159,160,161]
BAHA	C++/CUDA branch-aware basin hopping; 26-domain table; 109-file validation suite	solution/witness checks; failures documented (XOR-SAT, smooth landscapes, ChaCha20)	[154,155,162]
NitroSAT	physics-informed MaxSAT approximator (Python/C/Lua), V3 streaming	independent assignment verification (e.g. 923/923 clauses); Lean 4 proof files for the conditional lock	[156,163,164]

Artifact	What it is	Verification present	Ref
Casimir-SAT	physics-inspired SAT prototype (quantum-vacuum / Casimir forces)	benchmark scripts + SAT 2024 dataset; Zenodo DOI	[165]
Navokoj / SUTRA evidence bundle	CNF corpus, joined result CSVs, verifier records, manifest hashes	independent assignment checks; witness recount; client-verified artifact	[158]

Nothing here substitutes for independent replication (G1). But the artifacts make the correctness claims *checkable* rather than asserted: a reader can recompute every reported counter from the returned witness and re-run the oracle tests.

7.7 Resolution-depth regularisation: the observer bridge

Everything in this paper is indexed by the continuation set $\mathcal{W}$, and the monotone-coarsening property (Prop. 2.1) is stated in terms of shrinking it. When $\mathcal{W}$ is infinite and the process never halts, the natural index is not elapsed time but **resolution depth**: the level j of a refining partition tower, at which $\mathcal{W}$ has been coarsened to its resolution- j quotient X_j .

Substituting resolution depth for time in a survival-weighted observer gives a scalar that couples the quotient to the geometry of its completion. Let $c_j = |X_j|$ be the number of resolution- j quotient classes and let $1 - q$ be the observer’s survival probability per level. Define

$$O_q = \sum_{j \geq 0} c_j (1 - q)^j.$$

This is the generating function of the quotient-class counts, so its singularities are fixed by their growth.

Exponential growth. If $c_j \sim b^j$, then under the visual metric $d = 2^{-r}$ on the boundary of the discovery tree, $\dim(\partial T_{\text{nov}}) = \log_2 b$, and the sum converges iff $b(1 - q) < 1$. Hence

$$q_c = 1 - \frac{1}{b} = 1 - 2^{-\dim(\partial T_{\text{nov}})}.$$

The critical survival probability of a geometric observer is fixed by the dimension of the boundary.

Polynomial growth. If $c_j \sim (j + 1)^m$, the same sum is the object of the STOP residue theorem,

$$\text{FP}_{t \rightarrow 0} \sum_{k \geq 1} k^m e^{-t(k-1)} = \zeta(-m) + \frac{1}{m+1},$$

verified symbolically for $m = 0, \dots, 6$. The zeta residue is therefore the **sub-exponential case** of resolution-depth regularisation; exponential growth produces a dimension pole instead.

Status. This is a bridge, not a new minimality theorem, and it is deliberately labelled as elementary. The ingredients — geometric series and radius of convergence, box-counting

dimension on a tree boundary, Pringsheim’s theorem, and the residue identity above — are all standard. What is asserted is only that *resolution depth, not time, is the correct index* for a survival-weighted observer on the quotient’s completion, and the resulting critical scale. It is **not** an equivalence of categories: no functor is constructed here, and none is claimed. Nothing in this subsection bears on the transfer criterion (§5.4) or on hypotheses (A)–(C).

8. Synthesis

Layer	Category / topology	Discrete (cache, CDCL)	Continuous (relaxation, SOS)	Thermodynamic
Distinction	partitions $\mathcal{P}_j$ [28,29]	variables, cache keys	$z_i \in \mathbb{D}$, loss surface	microstates
Equivalence	$h_1 \sim_{\mathcal{W}} h_2$ [1,2,6,7]	task-invariant prompt class	$\nabla v = 0$ subspace	equal macroscopic observables
Quotient	terminal/coequalizer $\mathcal{M}^*$ [19]	minimal cache; UNSAT cores	relaxed classes with equal projections	coarse-grained macrostate
Geometry	$d = 2^{-k}$, Gromov boundary [24,25,31]	refinement/search trees	Fisher + cusp metric	replica ultrametric [92,93]
Compatibility	presheaf, $\tilde{H}^1$ [37,38,39]	implication/conflict graphs	prime gates	non-interfering modes
Dynamics	bonding maps q_j [28,29]	propagation, backjump, eviction	$\dot{x} = -g^{-1} \nabla F$ [32,34]	Langevin/annealing [45,46]
Selection	coherent profiles $\varprojlim X_j$	satisfying model	Lyapunov minimum	ground state
Witness	finite prefix, section	verification certificate	low-energy readout	local measurement

9. Conclusion

The unifying observation is a quotient identity:

$$H \xrightarrow{Q} \mathcal{M}^* \xrightarrow{\tilde{F}} \mathcal{G}.$$

State is the quotient of what happened by what can happen next. Every component is anchored: quotienting by future indistinguishability is Myhill-Nerode / causal-state / bisimulation / Galois-abstraction theory [1,2,6,7,15,16,17,18]; the generated geometry is the Gromov visual metric and the Fisher-Rao metric [24,25,31,32]; compatibility is sheaf cohomology [37,38,39]; the relaxation is self-concordant log-barrier plus natural gradient, Lyapunov-stable via LaSalle [32,34,47,48,59]; multiplicative gating is product-of-experts / multiplicative weights [52,54,55]; and the horizon is non-computable limits and undecidability [73,74,75,76,77]. Prime weighting and the Riemann lock remain conjectural, the

BAHA fracture jump is empirically supported with an informal derivation (§6.2, §7.5), and FUTCache and evidence compression are proposals without experiments. The programme is therefore not a new calculus but a *common language* for four existing calculi — and the open problems are exactly those flagged in §4, §5.2, §5.3, and Appendix D.

Forget when distinctions cease to matter. Decide when alternatives cease to differ.

Appendix A. Provenance map (claim → prior art)

Paper claim	Established grounding
Thm 2.1 universal factorization	coequalizer universal property [19,20]; minimal realization [21,22]; categorical DL [101,102,103]
Minimal future-sufficient state	Myhill–Nerode [1,2,3]; causal states [6,7]; PSRs [95,96,97]; bisimulation quotient [15,16]; MDP homomorphisms [98,99]
Redundancy = $\ker \phi$	best abstraction of Galois connection [17,18]
Monotone coarsening	monotone refinement of partitions / abstract domains [17,18,28]
Ultrametric from refinement	profinite/Gromov boundary [24,25,26]; dendrograms [27]
Discovery words	Mazurkiewicz traces / partial-order reduction [28,29,30]
Fisher metric & natural gradient	Chentsov, Amari, mirror descent [31,32,33,34,35]
Cusp/hyperbolic barrier	hyperbolic geometry, Poincaré embeddings [25,36,104,105]
Presheaf & $\check{H}^1$ obstruction	sheaf theory, contextuality, sheaf NNs [37,38,39,40,110,111,112,113,114]
Lyapunov dissipation	LaSalle invariance [47,59,60]; thermodynamic flows [119,120,121,122]
Log-barrier potential	self-concordant barriers [47,48]
Prime independence / anti-resonance	unique factorization, Kronecker–Weyl [71,72]
Multiplicative gating	product of experts [52]; multiplicative weights [54,55]; gradient conflict [53]
$1 - \sigma > \gamma$, Riemann lock	explicit formula, Bombieri–Vinogradov, random-matrix zeta zeros [64,65,67,68,69,70,123,124,125,126,127]
Lambert-W jump	Lambert W [61]; fold catastrophe [62,63]; annealing [45,46]
KV cache / context compression	rate-distortion, IB, H2O, StreamingLLM, KVQuant, Scissorhands, LLMingua [80,81,83,84,85,87,88,89,130,131,134]
Continuous SAT relaxation	mean-field/clustering, SDP, SOS, NeuroSAT/FourierSAT [41,42,43,44,49,50,51,136,138,139]
Computability horizon	Specker sequences, Gold, Rice, Pour-El–Richards, Siegelmann, Caro, Boche [73,74,75,77,78,144,145,146,147]

Paper claim	Established grounding
Glassy loss landscapes	DNN vs spin glass, stat-mech of learning, stiffness [115,116,117]
Constraint learning	constrained-optimization learning surveys [128]
Learned/continuous SAT	NeuroSAT, FourierSAT, neural SAT evaluation [136,137,138,139,140,141]
Long-context & efficiency	long-context survey, KV quantization, speculation [130,131,132,133,150]

Appendix B. Corrections to the original

1. **Metric inconsistency.** §3.4–3.5 uses inverse metric $g^{z\bar{z}} = \frac{1}{4}|z|^2(1 - |z|^2)^2$; §5.2 uses $g^{ii} = x_i(1 - x_i)$. They are different geometries. The Fisher/natural-gradient metric is the well-posed one for §5; the cusp metric is a separate design choice (§3.2).
2. **Prime mass.** $\sum_{c \leq K} W(p_c) = \Theta(K / \ln K)$, not $\sim K$ [72].
3. **“NP-hard $\equiv \check{H}^1 \neq 0$ ”** is false as stated; the paper already contains the counterexamples (2-SAT, Horn, GF(2)); we make the non-equivalence explicit [40].
4. **BAHA “Theorem 6.1”** is a heuristic derivation, not a theorem; the Fisher-information step is not rigorous [46,61,62]. The *method* is nevertheless empirically supported by the BAHA benchmark suite [154,155].
5. **BAHA pass rate.** The hosted article reports 23/26 (88%); the repository’s current table reports 22/26 (84%), with VRP, Set Cover, LABS, and HP Protein Folding listed as failures. We use the repository figure [162].

Appendix C. Citation policy

Each reference was admitted only if deleting it would remove a distinct theorem, construction, historical lineage, competing formulation, empirical result, or boundary on a claim; otherwise it was pruned. Grouped citations are used only where the grouped works jointly support one claim, and every clustered reference is either broken out with its specific contribution or removed. The bibliography is therefore load-bearing rather than exhaustive.

Appendix D. Contribution, novelty, and status of the bridges

D.1 The central universal property is not novel. Theorem 2.1 is the universal property of the quotient by an equivalence relation — the coequalizer of the kernel pair in **Set** [19,20] — specialized to “same behavior on the continuation set $\mathcal{W}$.” It coincides definitionally with the Nerode right congruence [1,2], the causal-state partition [6,7], bisimulation minimization [15,16], exact MDP aggregation [9,10], and Kalman/Willems minimal realization [21,22]; coalgebraically it is finality up to behavioral equivalence [13,14]. The proof is two lines because it is the definition of a quotient. Two further components are standard, not new:

- **$\mathcal{W}$ -parameterized coarsening.** The quotient depends on the test set $\mathcal{W}$, and $\mathcal{W}_1 \subseteq \mathcal{W}_2$ makes $\sim_{\mathcal{W}_1}$ coarser. This is exactly the coarsening used by Angluin’s observation table in active automata learning [5], by predictive state representations, which fix a chosen set of future tests [95], and by testing equivalences [153].

- **Monotone coarsening** (Prop. 2.1). Shrinking the test set can only merge classes. The forget/decide loop is a variant of the abstract/refine loop of counterexample-guided abstraction refinement, where added constraints shrink admissible behavior and coarsen the abstraction [152].

The “terminal” label carries no content: the quotient is terminal here only because morphisms are defined to run from finer to coarser representations; under the opposite convention it is initial. Nothing depends on the name.

D.2 Status of each bridge.

Claim	Status	Already given by	What would upgrade it
Universal factorization (Thm 2.1)	definitional consequence	coequalizer [19,20]; Nerode [1,2]; causal states [6,7]; bisimulation [15,16]; MDP aggregation [9,10]; minimal realization [21,22]; final coalgebra [13,14]	— (it is the definition of a quotient)
$\mathcal{W}$ -parameterized quotient	standard	observation table [5]; PSRs [95]; testing equivalence [153]	complexity of the coarsest task-preserving $\mathcal{W}$
Forget/decide loop (Prop. 2.1)	trivial + analogy	monotonicity under test restriction; CEGAR [152]	a coupled forget/decide theorem with a rate representation
Discrete ultrametric (Thm 3.1)	consequence of the construction	profinite/Gromov boundary [24,25,26]; behavioral pseudometrics [11]	theorem: every computational distinguishability is such a limit
Tree/Gromov identification	consequence	Gromov visual metric [24]	—
Fisher / natural-gradient flow	consequence	Amari [31,32]; mirror descent [34,35]	—
Transfer criterion (§5.4)	criterion + partial theorem: convexity clause provable on Horn/submodular, parity obstruction known; full characterization open	discrete convexity, Schaefer dichotomy, Lovász extension, median graphs [166–174]	the necessary-and-sufficient characterization
Chebyshev recovery rate (§5.5)	proved (quadratic basin): $O(\sqrt{\kappa} \log(D/\rho_0))$ identification; filtered method required	Chebyshev minimax [178]; accelerated methods [176]; Chebyshev semi-iteration / CG [175,177]	nonlinear-basin version; application to §5.1’s unfiltered flow

Claim	Status	Already given by	What would upgrade it
Conditional quotient realization (Thm 5.2)	proved conditional on (A) ; (B), (C) are structural assumptions	Conley Morse decomposition [179]; Chebyshev rate [175–178]	(A) for a concrete class; submodular realizes only the coarsest quotient [180]
Worked non-convex instances (§5.7)	proved : 1-D A1 holds; 2-D A1 fails above $\sigma^2/2$ via saddle-node	Morse classification [180]; saddle-node bifurcation [181]	a rich-quotient non-convex class with no surplus index-0 components
BAHA navigation / A1' (§5.8)	operational : fracture $\equiv$ fold; Lambert-W branches $\equiv$ fold continuations; branch jumping $\equiv$ Morse navigation	BAHA [154,155,162]; Lambert W [61]	A1' proved for a class; control of detector false positives
SOS-certified realization (Lem. 5.1)	proved, conditional on the certificate : A1 holds iff there are no unlabeled index-0 critical points; certificate degree is the complexity measure	Positivstellensatz [182,183,184]; SOS hierarchy [55,56]	a bounded-degree certificate for a rich quotient
Inverted Poincaré well; quadratic quenching (Thm 3.2–3.3)	new construction (proved); bridge is analogy	hyperbolic geometry [25,36]	transfer theorem: SAT correctness/approximation guarantee
Presheaf / Čech machinery (§4)	consequence	sheaf theory [37,40,41]	—
Cohomology $\leftrightarrow$ hardness	analogy (paper disclaims complexity content)	CSP algebra [40]; sheaf contextuality [37]	lower bound on $\check{H}^1$ dimension vs. runtime for a family
Lyapunov dissipation (Thm 5.1)	consequence	LaSalle [59,60]; log-barrier [47,48]	—
Prime anti-resonance (§5.2)	conjecture	Kronecker-Weyl [71] gives independence only	proof or counterexample
$1 - \sigma > \gamma$; Riemann lock (§5.3)	heuristic / conjecture	explicit formula [67]; Bombieri-Vinogradov [64]	prove the coupling, or refute it
Multiplicative direction preservation (Prop. 6.1)	trivial consequence	product of experts [52]; barrier methods [47]	—
BAHA / Lambert-W jump (§6.2)	empirically supported ; derivation heuristic; branch completeness conditional	catastrophe theory [62]; annealing [45]; Lambert W [61]; BAHA benchmarks [154,155]	independent SOTA-baseline comparison; rigor for the Laplace-horizon step

Claim	Status	Already given by	What would upgrade it
KV / context quotient (§7.1–7.2)	reframing of known methods	rate-distortion / IB [8,81]; bisimulation metrics [11]; H2O [84]; StreamingLLM [85]; LLMingua [87]	estimator + regret bound + experiments
Computability horizon (Thm 7.1)	consequence	Turing [76]; Specker [73]; Gold [75]	—

D.3 What would make the programme a contribution.

1. **A transfer theorem.** Prove that for a class of discrete systems the continuous relaxation’s quotient is isomorphic (or bi-Lipschitz) to the discrete quotient, with a computable constant. Without this, “generated geometry” is an analogy. The structural form of the hypotheses is stated in §5.4: an exact convexity preserved by the relaxation, separation of quotient classes by the observables, and uniform conditioning of the basins — with parity as the canonical obstruction.
2. **A coupled forget/decide theorem.** Define a process interleaving quotienting over $\mathcal{W}_t$ with constraint propagation, and prove soundness and a rate at which $\mathcal{W}_t$ shrinks. Prop. 2.1 alone is trivial.
3. **Prime weights.** Either prove a cancellation-avoidance bound, or report the likely negative result: real clause potentials are not rescued from harmonic cancellation by $\mathbb{Q}$ -linear independence of $\log p$ alone, since the relevant obstruction is conditioning of the weighted clause operator.
4. **Cohomology.** Exhibit a family with a lower bound on $\dim \check{H}^1$ and a matching runtime separation. Otherwise the machinery is decoration.
5. **Comparative, bias-controlled, replicated experiments.** The correctness evidence is already of the right kind — witnesses checked against the original instances by separate verifiers [154,155,156,158] — so the open question is not validity but scope. What is missing is (i) apples-to-apples comparison against modern CDCL / Survey Propagation / MaxSAT solvers and against H2O / StreamingLLM / LLMingua, (ii) control for benchmark-selection bias,
- (iii) generalization beyond the tested distributions, and (iv) independent replication by a third party. Provenance determines independence; verification determines validity; only (i)–(iv) concern provenance and scope.
6. **The only possibly-new mathematical object** is the interaction-aware, $\mathcal{W}$ -parameterized terminal quotient as an *algorithm-design criterion* (does a compressed state preserve the quotient?), not the universal property itself. That is a design principle, and it should be claimed as such.

Appendix E. Outlook: certified behavioral minima, interpretability, and alignment

The transfer machinery has an obvious target beyond constraint solvers: the loss landscape of a trained network. Parameters play the role of histories, an explicit behavioral test family $\mathcal{W}$ plays the role of the continuation set, and two parameter settings are behaviorally equivalent when they answer every probe in $\mathcal{W}$ identically. Writing $Q_{\mathcal{W}}$ for the induced behavioral quotient and $\mathcal{M}_{\mathcal{W}}^* = X / \sim_{\mathcal{W}}$, a *mechanistic certificate* is the following pair of facts.

1. **Attractor completeness.** $\text{Crit}_0(L)/G = \{C_1, \dots, C_k\}$ is finite and isolated modulo a symmetry group G whose orbits preserve the realized function — certified by Lemma 5.1 in a symmetry-reduced or equivariant Morse–Bott formulation [190,191].
2. **Behavioral correspondence.** $\pi : \{C_1, \dots, C_k\} \xrightarrow{\cong} \mathcal{M}_{\mathcal{W}}^*$, i.e. the attractors biject onto the $\mathcal{W}$ -classes.

The intended object is the commuting diagram

$$\begin{array}{ccc}
 \text{Crit}_0(L)/G & \xrightarrow{\pi} & \mathcal{M}_{\mathcal{W}}^* \\
 \uparrow & & \uparrow Q_{\mathcal{W}} \\
 \text{training dynamics} & & \text{observable behavior}
 \end{array}$$

What this proves, and only this.

If the attracting components of a model’s training dynamics can be certified to correspond exactly to a specified behavioral quotient, then learning acquires a mechanically checkable behavioral certificate: every reachable attractor realizes a known behavior class, and no additional attracting behavior exists within the certified domain.

Equivalently: *there are no attracting behavioral modes outside the certified taxonomy* — within the modeled parameter region and relative to $\mathcal{W}$. This is not “we proved it learned the right thing.” It says exactly what has been proved and exactly what remains dependent on the specification $\mathcal{W}$.

Why “relative to $\mathcal{W}$ ” is everything. The quotient is monotone in the test family,

$$\mathcal{W}_1 \subset \mathcal{W}_2 \implies Q_{\mathcal{W}_2} \text{ refines } Q_{\mathcal{W}_1},$$

so a certificate is only as strong as its probes. If $\mathcal{W}$ does not test deception, backdoors, or distribution shift, two parameter settings can be $\mathcal{W}$ -equivalent while differing catastrophically outside $\mathcal{W}$. Alignment is therefore partly a **test-completeness** problem: the landscape certificate proves there are no extra attractors *relative to the partition you specified*; it cannot prove the specification captures right behavior. That is the specification gap [193,194], and it is untouched by any amount of SOS degree.

Symmetry. Individual parameter-space minima are the wrong objects: hidden-unit permutation and rescaling generate large equivalence families with identical realized functions [190,191]. The census must be taken over $\text{Crit}_0(L)/G$, or over connected attracting components / function-space equivalence classes, rather than by counting parameter-space critical points.

BAHA as a re-verification trigger. If training crosses a bifurcation at which a new index-0 component appears, then $|\text{Crit}_0(L)/G| : k \rightarrow k + 1$. The certificate remains valid iff the new attractor maps to an already certified $\mathcal{W}$ -class; if it does not, the certificate has just failed. A bifurcation (fracture) detector is thus a **trigger for re-verification**, not a licence to assume training stayed inside the certified regime. This is the alignment reading of §5.8.

The dangerous object is not a spurious minimum. It is an attractor the behavioral quotient failed to name.

What remains, precisely. Three gaps survive the corrections. (i) *Polynomial form*: Lemma 5.1 needs a polynomial F ; the certificate applies to polynomial-activation networks, smooth activations after polynomial approximation, or a certified surrogate. (ii) *The behavioral map*: the certificate needs a positive bound relating parameter distance to behavioral difference

on $\mathcal{W}$ [192]; without it, certified attractors are not certified semantics. (iii) *The specification gap*, above.

With these named, the machinery still delivers a real, checkable property: **certified behavioral homogeneity of basins**. If $B(x_i)$ is certified to lie in a parameter ball of radius r , the network is parameter-Lipschitz with constant ℓ there, and the probes are $\ell_{\mathcal{W}}$ -Lipschitz, then every parameter in $B(x_i)$ agrees with x_i on every probe to within $\ell \ell_{\mathcal{W}} r$. That is the network analogue of a quotient class, and a genuine safety property — behavioral invariance under retraining noise, fine-tuning within a basin, or bounded parameter perturbation.

If $\mathcal{W}$ is chosen to be a circuit-level probe set [185], the quotient classes are the circuits, and a certified finite attractor census becomes a *certified circuit census*; probing superposition structure [186] gives the feature-basis variant. Landscape structure makes this plausible rather than hopeless: the surface is highly non-convex with many near-flat directions [187,188], but minima are connected by low-loss paths [189], which is what a finitary Morse-Bott census requires.

References

1. Myhill, J. (1957). *Finite automata and the representation of events*. WADC Technical Report 57-624.
2. Nerode, A. (1958). Linear automaton transformations. *Proc. AMS* 9(4), 541-544.
3. Hopcroft, J. E. (1971). An $n \log n$ algorithm for minimizing states in a finite automaton. In *Theory of Machines and Computations*, 189-196.
4. Paige, R., & Tarjan, R. E. (1987). Three partition refinement algorithms. *SIAM J. Comput.* 16(6), 973-989.
5. Angluin, D. (1987). Learning regular sets from queries and counterexamples. *Inf. Comput.* 75(2), 87-106.
6. Crutchfield, J. P. (1994). The calculi of emergence: computation, dynamics and induction. *Physica D* 75, 11-54.
7. Shalizi, C. R., & Crutchfield, J. P. (2001). Computational mechanics: pattern and prediction, structure and randomness. *J. Stat. Phys.* 104, 817-879.
8. Bialek, W., Nemenman, I., & Tishby, N. (2001). Predictability, complexity, and learning. *Neural Comput.* 13(11), 2409-2463.
9. Givan, R., Dean, T., & Greig, M. (2003). Equivalence notions and model minimization in Markov decision processes. *Artificial Intelligence* 147(1-2), 163-223.
10. Li, L., Walsh, T. J., & Littman, M. L. (2006). Towards a unified theory of state abstraction for MDPs. *ISAIM*.
11. Ferns, N., Panangaden, P., & Precup, D. (2004). Metrics for finite Markov decision processes. *UAI*, 162-169.
12. Abel, D., Hershkowitz, D., & Littman, M. L. (2016). Near-optimal approximate state abstraction. *UAI*.
13. Rutten, J. J. M. M. (2000). Universal coalgebra: a theory of systems. *Theoret. Comput. Sci.* 249(1), 3-80.
14. Jacobs, B. (2016). *Introduction to Coalgebra: Towards Mathematics of States and Observation*. Cambridge University Press.
15. Milner, R. (1989). *Communication and Concurrency*. Prentice Hall.
16. Park, D. (1981). Concurrency and automata on infinite sequences. *LNCS* 104, 167-183.
17. Cousot, P., & Cousot, R. (1977). Abstract interpretation: a unified lattice model for static analysis. *POPL*, 238-252.
18. Cousot, P., & Cousot, R. (1979). Systematic design of program analysis frameworks. *POPL*, 269-282.
19. Mac Lane, S. (1998). *Categories for the Working Mathematician* (2nd ed.). Springer.

20. Adámek, J., Herrlich, H., & Strecker, G. E. (1990). *Abstract and Concrete Categories*. Wiley.
21. Kalman, R. E. (1963). Mathematical description of linear dynamical systems. *SIAM J. Control* 1(2), 152–192.
22. Willems, J. C. (1991). Paradigms and puzzles in the theory of dynamical systems. *IEEE Trans. Autom. Control* 36(3), 259–294.
23. Krohn, K., & Rhodes, J. (1965). Algebraic theory of machines I. *Trans. AMS* 116, 450–464.
24. Gromov, M. (1987). Hyperbolic groups. In *Essays in Group Theory*, Springer, 75–263.
25. Bridson, M. R., & Haefliger, A. (1999). *Metric Spaces of Non-Positive Curvature*. Springer.
26. Ribes, L., & Zalesskii, P. (2010). *Profinite Groups* (2nd ed.). Springer.
27. Carlsson, G., & Mémoli, F. (2010). Characterization, stability and convergence of hierarchical clustering methods. *JMLR* 11, 1425–1470.
28. Mazurkiewicz, A. (1987). Trace theory. *LNCS* 266, 279–324.
29. Godefroid, P. (1996). *Partial-Order Methods for the Verification of Concurrent Systems*. Springer.
30. Winskel, G. (1987). Event structures. *LNCS* 266, 325–392.
31. Chentsov, N. N. (1982). *Statistical Decision Rules and Optimal Inference*. AMS.
32. Amari, S. (1998). Natural gradient works efficiently in learning. *Neural Comput.* 10(2), 251–276.
33. Amari, S., & Nagaoka, H. (2000). *Methods of Information Geometry*. AMS.
34. Raskutti, G., & Mukherjee, S. (2015). The information geometry of mirror descent. *IEEE Trans. Inf. Theory* 61(7), 3991–4008.
35. Beck, A., & Teboulle, M. (2003). Mirror descent and nonlinear projected subgradient methods. *Oper. Res. Lett.* 31(3), 167–175.
36. Nickel, M., & Kiela, D. (2017). Poincaré embeddings for learning hierarchical representations. *NeurIPS*.
37. Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure of non-locality and contextuality. *New J. Phys.* 13, 113036.
38. Ghrist, R. (2014). *Elementary Applied Topology*. CreateSpace.
39. Hansen, J., & Ghrist, R. (2019). Toward a spectral theory of cellular sheaves. *J. Appl. Comput. Topol.* 3, 315–358.
40. Schaefer, T. J. (1978). The complexity of satisfiability problems. *STOC*, 216–226. (see also Aspvall, Plass & Tarjan, *Inf. Process. Lett.* 8(3), 121–123, 1979.)
41. Mézard, M., Parisi, G., & Zecchina, R. (2002). Analytic and algorithmic solution of random satisfiability problems. *Science* 297(5582), 812–815.
42. Mézard, M., & Montanari, A. (2009). *Information, Physics, and Computation*. Oxford University Press.
43. Monasson, R., & Zecchina, R. (1996). Entropy of the K -satisfiability problem. *Phys. Rev. Lett.* 76(21), 3881–3885.
44. Krzakala, F., Montanari, A., Ricci-Tersenghi, F., Semerjian, G., & Zdeborová, L. (2007). Gibbs states and the set of solutions of random constraint satisfaction problems. *PNAS* 104(25), 10318–10323.
45. Kirkpatrick, S., Gelatt, C. D., & Vecchi, M. P. (1983). Optimization by simulated annealing. *Science* 220(4598), 671–680.
46. Neal, R. M. (2001). Annealed importance sampling. *Stat. Comput.* 11, 125–139.
47. Nesterov, Y., & Nemirovskii, A. (1994). *Interior-Point Polynomial Algorithms in Convex Programming*. SIAM.
48. Nesterov, Y. (2004). *Introductory Lectures on Convex Optimization*. Springer.
49. Goemans, M. X., & Williamson, D. P. (1995). Improved approximation algorithms for maximum cut and satisfiability problems using semidefinite programming. *JACM* 42(6), 1115–1145.
50. Lasserre, J. B. (2001). Global optimization with polynomials and the problem of moments.

- SIAM J. Optim.* 11(3), 796–817.
51. Parrilo, P. A. (2003). Semidefinite programming relaxations for semialgebraic problems. *Math. Program.* 96, 293–320.
 52. Hinton, G. E. (2002). Training products of experts by minimizing contrastive divergence. *Neural Comput.* 14(8), 1771–1800.
 53. Yu, T., Kumar, S., Gupta, A., Levine, S., Hausman, K., & Finn, C. (2020). Gradient surgery for multi-task learning. *NeurIPS*.
 54. Kivinen, J., & Warmuth, M. K. (1997). Exponentiated gradient versus gradient descent for linear predictors. *Inf. Comput.* 132(1), 1–63.
 55. Arora, S., Hazan, E., & Kale, S. (2012). The multiplicative weights update method: a meta-algorithm and applications. *Theory Comput.* 8, 121–164.
 56. Fiedler, M. (1973). Algebraic connectivity of graphs. *Czechoslovak Math. J.* 23(2), 298–305.
 57. Cheeger, J. (1970). A lower bound for the smallest eigenvalue of the Laplacian. In *Problems in Analysis*, Princeton, 195–199.
 58. Levin, D. A., & Peres, Y. (2017). *Markov Chains and Mixing Times* (2nd ed.). AMS.
 59. LaSalle, J. P. (1960). Some extensions of Liapunov’s second method. *IRE Trans. Circuit Theory* 7(4), 520–527.
 60. Khalil, H. K. (2002). *Nonlinear Systems* (3rd ed.). Prentice Hall.
 61. Corless, R. M., Gonnet, G. H., Hare, D. E. G., Jeffrey, D. J., & Knuth, D. E. (1996). On the Lambert W function. *Adv. Comput. Math.* 5, 329–359.
 62. Thom, R. (1972). *Stabilité Structurelle et Morphogénèse*. Benjamin.
 63. Zeeman, E. C. (1977). *Catastrophe Theory: Selected Papers 1972–1977*. Addison-Wesley.
 64. Bombieri, E. (1965). On the large sieve. *Mathematika* 12(2), 201–225.
 65. Vinogradov, A. I. (1965). On the density hypothesis for Dirichlet L -series. *Izv. Akad. Nauk SSSR* 29, 903–934.
 66. Montgomery, H. L., & Vaughan, R. C. (1974). Hilbert’s inequality. *J. London Math. Soc.* 2(1), 73–82.
 67. Titchmarsh, E. C. (1986). *The Theory of the Riemann Zeta-Function* (2nd ed.). Oxford University Press.
 68. Ingham, A. E. (1937). On the difference between consecutive primes. *Quart. J. Math.* 8, 255–266.
 69. Connes, A. (1999). Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. *Selecta Math.* 5(1), 29–106.
 70. Berry, M. V., & Keating, J. P. (1999). The Riemann zeros and eigenvalue asymptotics. *SIAM Rev.* 41(2), 236–266.
 71. Weyl, H. (1916). Über die Gleichverteilung von Zahlen mod. Eins. *Math. Ann.* 77, 313–352.
 72. Hardy, G. H., & Wright, E. M. (2008). *An Introduction to the Theory of Numbers* (6th ed.). Oxford University Press.
 73. Specker, E. (1949). Nicht konstruktiv beweisbare Sätze der Analysis. *J. Symbolic Logic* 14(3), 145–158.
 74. Rice, H. G. (1953). Classes of recursively enumerable sets and their decision problems. *Trans. AMS* 74(2), 358–366.
 75. Gold, E. M. (1967). Language identification in the limit. *Inf. Control* 10(5), 447–474.
 76. Turing, A. M. (1936). On computable numbers, with an application to the Entscheidungsproblem. *Proc. London Math. Soc.* 2(42), 230–265.
 77. Pour-El, M. B., & Richards, J. I. (1989). *Computability in Analysis and Physics*. Springer.
 78. Weihrauch, K. (2000). *Computable Analysis: An Introduction*. Springer.
 79. Li, M., & Vitányi, P. (2008). *An Introduction to Kolmogorov Complexity and Its Applications* (3rd ed.). Springer.
 80. Tishby, N., Pereira, F. C., & Bialek, W. (1999). The information bottleneck method. *Allerton*.

81. Shannon, C. E. (1959). Coding theorems for a discrete source with a fidelity criterion. *IRE Nat. Conv. Rec.* 7(4), 142–163.
82. Cover, T. M., & Thomas, J. A. (2006). *Elements of Information Theory* (2nd ed.). Wiley.
83. Kwon, W., Li, Z., Zhuang, S., Sheng, Y., Zheng, L., Yu, C. H., Gonzalez, J. E., Zhang, H., & Stoica, I. (2023). Efficient memory management for large language model serving with PagedAttention. *SOSP*.
84. Zhang, Z., Sheng, Y., Zhou, T., Chen, T., Zheng, L., Cai, R., Song, Z., Tian, Y., Ré, C., Barrett, C., Wang, Z., & Chen, B. (2023). H₂O: Heavy-hitter oracle for efficient generative inference of large language models. *NeurIPS*.
85. Xiao, G., Tian, Y., Chen, B., Han, S., & Lewis, M. (2023). Efficient streaming language models with attention sinks. *ICLR 2024*.
86. Beltagy, I., Peters, M. E., & Cohan, A. (2020). Longformer: The long-document transformer. *arXiv:2004.05150*.
87. Jiang, H. Z., Wu, Q., Lin, C.-Y., Yang, Y., & Qiu, L. (2023). LLMingua: Compressing prompts for accelerated inference of large language models. *EMNLP*.
88. Pan, Z., Wu, Q., Jiang, H., Xia, M., Luo, X., Zhang, J., Lin, Q., Rühle, V., Yang, Y., Lin, C.-Y., Zhao, H. V., Qiu, L., & Zhang, D. (2024). LLMingua-2: Data distillation for efficient and faithful task-agnostic prompt compression. *ACL Findings*.
89. Mu, J., Li, X. L., & Goodman, N. (2023). Learning to compress prompts with gist tokens. *NeurIPS*.
90. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS*.
91. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *NeurIPS*.
92. Parisi, G. (1980). The order parameter for spin glasses: a function on the interval 0–1. *J. Phys. A* 13, 1101–1112.
93. Mézard, M., Parisi, G., Sourlas, N., Toulouse, G., & Virasoro, M. (1987). *Spin Glass Theory and Beyond*. World Scientific.
94. Connes, A., & Bost, J.-B. (1995). Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory. *Selecta Math.* 1(3), 411–457.
95. Littman, M. L., Sutton, R. S., & Singh, S. (2002). Predictive representations of state. *NeurIPS*.
96. Kaelbling, L. P., Littman, M. L., & Cassandra, A. R. (1998). Planning and acting in partially observable stochastic domains. *Artificial Intelligence* 101(1–2), 99–134.
97. Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press.
98. Ravindran, B., & Barto, A. G. (2003). SMDP homomorphisms: an algebraic approach to abstraction in semi-Markov decision processes. *IJCAI*.
99. Biza, O., Platt, R., van de Meent, J.-W., & Wong, L. (2020). Online abstraction with MDP homomorphisms for deep reinforcement learning. *AAMAS*.
100. Dorsch, U., Milius, S., & Schröder, L. (2017). Efficient coalgebraic partition refinement. *CONCUR*; arXiv:1705.02882.
101. Gavranović, B., Lessard, P., Dudzik, A., von Glehn, T., Araújo, J. G. M., & Veličković, P. (2024). Position: Categorical deep learning is an algebraic theory of all architectures. *ICML*; arXiv:2402.15332.
102. Fong, B., & Spivak, D. I. (2019). *An Invitation to Applied Category Theory*. Cambridge University Press.
103. Spivak, D. I. (2014). *Category Theory for the Sciences*. MIT Press.
104. Peng, W., Varanka, T., Mostafa, A., Shi, H., & Zhao, G. (2022). Hyperbolic deep neural networks: a survey. *IEEE TPAMI* 44(12), 10023–10044.
105. Sala, F., De Sa, C., Gu, A., & Ré, C. (2018). Representation tradeoffs for hyperbolic embeddings. *ICML*.
106. Zhang, L., Naitzat, G., & Lim, L.-H. (2018). Tropical geometry of deep neural networks.

- ICML; arXiv:1805.07091.
107. Peyré, G., & Cuturi, M. (2019). Computational optimal transport. *Found. Trends Mach. Learn.* 11(5–6), 355–607.
 108. Cohen-Addad, V., Das, A., Karthik, C. S., & Mathieu, C. (2020). On efficient low distortion ultrametric embedding. *ICML*.
 109. Bauer, M., Mémoli, F., Needham, T., & Nishino, M. (2025). The Z-Gromov-Wasserstein distance. *JMLR* 26; arXiv:2408.08233.
 110. Bodnar, C., Di Giovanni, F., Chamberlain, B. P., Liò, P., & Bronstein, M. M. (2022). Neural sheaf diffusion: a topological perspective on heterophily and oversmoothing in GNNs. *NeurIPS*; arXiv:2202.04579.
 111. Hansen, J., & Gebhart, T. (2020). Sheaf neural networks. arXiv:2012.06333.
 112. Barbero, F., Bodnar, C., Sáez de Ocáriz Borde, H., Bronstein, M., Veličković, P., & Liò, P. (2022). Sheaf neural networks with connection Laplacians. *PMLR (Topological, Algebraic and Geometric Learning Workshops)*.
 113. Ayzenberg, A., Gebhart, T., Magai, G., & Solomadin, G. (2025). Sheaf theory: from deep geometry to deep learning. arXiv:2502.15476.
 114. Bach, E. (1999). Sheaf cohomology is #P-hard. *J. Symbolic Computation* 27(4), 429–433.
 115. Baity-Jesi, M., Sagun, L., Geiger, M., Spigler, S., Ben Arous, G., Cammarota, C., LeCun, Y., Wyart, M., & Biroli, G. (2019). Comparing dynamics: deep neural networks versus glassy systems. *J. Stat. Mech.* 124013; arXiv:1803.06969.
 116. Bahri, Y., Kadmon, J., Pennington, J., Schoenholz, S. S., Sohl-Dickstein, J., & Ganguli, S. (2020). Statistical mechanics of deep learning. *Annu. Rev. Condens. Matter Phys.* 11, 501–528.
 117. Fort, S., Nowak, A., Jastrzębski, S., & Narayanan, S. (2021). Stiffness: a new perspective on generalization in neural networks. arXiv:1901.09491.
 118. Fatkhullin, I., Sokolov, I., Gorbunov, E., & Richtárik, P. (2024). Taming nonconvex stochastic mirror descent with general Bregman divergence. *ICML*.
 119. Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM J. Res. Dev.* 5(3), 183–191.
 120. Bennett, C. H. (1973). Logical reversibility of computation. *IBM J. Res. Dev.* 17(6), 525–532.
 121. Friston, K. (2010). The free-energy principle: a unified brain theory? *Nat. Rev. Neurosci.* 11, 127–138.
 122. Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics of prediction. *Phys. Rev. Lett.* 109, 120604.
 123. Bogomolny, E. B., & Keating, J. P. (1995). Random matrix theory and the Riemann zeros I: three- and four-point correlations. *Nonlinearity* 8(6), 1115–1131.
 124. Keating, J. P., & Snaith, N. C. (2000). Random matrix theory and $\zeta(1/2 + it)$. *Comm. Math. Phys.* 214, 57–89.
 125. Granville, A., & Soundararajan, K. (2007). Large deviations of the Riemann zeta function. *Acta Math.* 198, 1–47.
 126. Maynard, J. (2015). Small gaps between primes. *Ann. of Math.* 181(1), 383–413.
 127. Matomäki, K., & Radziwiłł, M. (2016). Multiplicative functions in short intervals. *Ann. of Math.* 183(3), 1015–1056.
 128. Kotary, J., Fioretto, F., Van Hentenryck, P., & Wilder, B. (2021). End-to-end constrained optimization learning: a survey. *IJCAI*.
 129. Du, Y., Li, S., Tenenbaum, J., & Mordatch, I. (2020). Compositional visual generation with energy based models. *NeurIPS*.
 130. Hooper, C., Kim, S., Mohammadzadeh, H., Mahoney, M. W., Shao, Y. S., Keutzer, K., & Gholami, A. (2024). KVQuant: towards 10 million context length LLM inference with KV cache quantization. *NeurIPS*; arXiv:2401.18079.
 131. Liu, Z., Desai, A., Liao, F., Wang, W., Xie, V., Xu, Z., Kyrillidis, A., & Shrivastava, A. (2023). Scissorhands: exploiting the persistence of importance hypothesis for LLM KV cache compression at test time. *NeurIPS*; arXiv:2305.17118.

132. Xia, H., Yang, Z., Dong, Q., et al. (2024). Unlocking efficiency in large language model inference: a comprehensive survey of speculative decoding. *ACL Findings*.
133. Huang, Y., Xu, J., Lai, J., Jiang, Z., Chen, T., Li, Z., Yao, Y., Ma, X., Yang, L., Chen, H., Li, S., & Zhao, P. (2023). Advancing transformer architecture in long-context large language models: a comprehensive survey. arXiv:2311.12351.
134. Li, Z., Liu, Y., Su, Y., & Collier, N. (2024). Prompt compression for large language models: a survey. *NAACL 2025*; arXiv:2410.12388.
135. Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, M., & Wang, H. (2024). Retrieval-augmented generation for large language models: a survey. arXiv:2312.10997.
136. Selsam, D., Lamm, M., Bünz, B., Liang, P., de Moura, L., & Dill, D. L. (2019). Learning a SAT solver from single-bit supervision. *ICLR*; arXiv:1802.03685.
137. Mojžišek, D., Hůla, J., Li, Z., Zhou, Z., & Janota, M. (2025). Neural approaches to SAT solving: design choices and interpretability. arXiv:2504.01173.
138. Sinjorgo, L., & Sotirov, R. (2023). On solving MAX-SAT using sum of squares. *INFORMS J. Comput.*; arXiv:2302.06931.
139. Kyrillidis, A., Shrivastava, A., Vardi, M. Y., & Zhang, Z. (2019). FourierSAT: a Fourier expansion-based algebraic framework for solving hybrid Boolean constraints. *AAAI*; arXiv:1912.01032.
140. Braunstein, A., Mézard, M., & Zecchina, R. (2005). Survey propagation: an algorithm for satisfiability. *Random Struct. Algorithms* 27(2), 201–226.
141. Marino, R., Parisi, G., & Ricci-Tersenghi, F. (2016). The backtracking survey propagation algorithm for solving random K-SAT problems. *Nat. Commun.* 7, 12996.
142. Hahn, M. (2020). Theoretical limitations of self-attention in neural sequence models. *TACL* 8, 156–171; arXiv:1906.06755.
143. Weiss, G., Goldberg, Y., & Yahav, E. (2021). Thinking like transformers. *ICML*; arXiv:2106.06981.
144. Siegelmann, H. T., & Sontag, E. D. (1995). On the computational power of neural nets. *J. Comput. Syst. Sci.* 50(1), 132–150.
145. Siegelmann, H. T. (1995). Computation beyond the Turing limit. *Science* 268(5210), 545–548.
146. Boche, H., Fojtik, V., Fono, A., & Kutyniok, G. (2024). Computability of classification and deep learning: from theoretical limits to practical feasibility through quantization. arXiv:2408.06212.
147. Caro, M. C. (2023). From undecidability of non-triviality and finiteness to undecidability of learnability. *Int. J. Approx. Reasoning*; arXiv:2106.01382.
148. Blum, L., Shub, M., & Smale, S. (1989). On a theory of computation and complexity over the real numbers. *Bull. Amer. Math. Soc.* 21(1), 1–46.
149. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
150. Tandon, A., Dalal, K., Li, X., Kocejka, D., Rød, M., Buchanan, S., Wang, X., Leskovec, J., Koyejo, S., Hashimoto, T., Guestrin, C., McCaleb, J., Choi, Y., & Sun, Y. (2025). End-to-end test-time training for long context. arXiv:2512.23675.
151. Bronstein, M. M., Bruna, J., Cohen, T., & Veličković, P. (2021). Geometric deep learning: grids, groups, graphs, geodesics, and gauges. arXiv:2104.13478.
152. Clarke, E., Grumberg, O., Jha, S., Lu, Y., & Veith, H. (2003). Counterexample-guided abstraction refinement for symbolic model checking. *JACM* 50(5), 752–794.
153. De Nicola, R., & Hennessy, M. C. B. (1984). Testing equivalences for processes. *Theoret. Comput. Sci.* 34(1–2), 83–133.
154. Iyer, S. (2026a). Branch Aware Exact Basin Hopping (BAHA): thermodynamic fracture detection with branch jumping. <https://sethuiyer.github.io/baha/>
155. Iyer, S. (2026b). Multiplicative calculus for hardness detection and branch-aware optimization: a computational framework for detecting phase transitions via non-integrable log-derivatives. Zenodo. <https://doi.org/10.5281/zenodo.18373732>
156. Iyer, S. (2026c). NitroSAT: solving MaxSAT through number theory, statistical mechan-

- ics, and persistent homology. <https://sethuiyer.codeberg.page/NitroSAT/>
157. Iyer, S. (2026d). NitroSAT V3 witness-recount and clause-level assignment verification protocol. Navokoj evidence workspace. (Independent witness recount; divergent counters are rejected, not propagated.)
 158. Iyer, S. (2026e). ShunyaBar evidence bundle: Navokoj / NitroSAT / SUTRA reproducibility artifacts, CNF corpus, and independent assignment checks. Hugging Face: sethuiyer/shunyabar-evidence-v1. <https://huggingface.co/buckets/sethuiyer/shunyabar-evidence-v1>
 159. Iyer, S. (2026f). FUTCache: a metric visited-set and sufficient-state compressor for future-novelty decisions. GitHub: sethuiyer/FUTCache. <https://github.com/sethuiyer/FUTCache>
 160. Iyer, S. (2026g). Empirical scaling benchmark on 1,000,000 real Alibaba Cloud traces, and geometric recurrence in CDCL proof search. FUTCache documentation.
 161. Iyer, S. (2026h). Negative result: geometric MDL is not semantic safety. FUTCache documentation.
 162. Iyer, S. (2026i). BAHa: branch-aware basin hopping (C++/CUDA, 26-domain validation suite). GitHub: sethuiyer/baha. <https://github.com/sethuiyer/baha>
 163. Iyer, S. (2026j). NitroSAT: physics-informed MaxSAT approximator. GitHub: sethuiyer/NitroSAT. <https://github.com/sethuiyer/NitroSAT>
 164. Iyer, S. (2026k). NitroSAT formal proofs in Lean 4 / Mathlib: ConditionalAsymptoticLock and NitroSATProofs. GitHub: sethuiyer/NitroSAT. (No sorry, axiom, or admit; hard analytic/spectral inputs are named hypotheses.)
 165. Iyer, S. (2026l). Casimir-SAT: solving SAT with quantum vacuum dynamics. GitHub: sethuiyer/casimir-sat-solver; Zenodo DOI 10.5281/zenodo.17394164.
 166. Schaefer, T. J. (1978). The complexity of satisfiability problems. *STOC*, 216–226.
 167. Jeavons, P., Cohen, D., & Gyssens, M. (1997). Closure properties of constraints. *JACM* 44(4), 527–548.
 168. Feder, T., & Vardi, M. Y. (1998). The computational structure of monotone monadic SNP and constraint satisfaction: a study through Datalog and group theory. *SIAM J. Comput.* 28(1), 57–104.
 169. Lovász, L. (1983). Submodular functions and convexity. In *Mathematical Programming: The State of the Art*, Springer, 235–257.
 170. Murota, K. (2003). *Discrete Convex Analysis*. SIAM.
 171. Chepoi, V. (2000). Graphs of some CAT(0) complexes. *Adv. Appl. Math.* 24(2), 125–179.
 172. Grigoriev, D. (2001). Linear lower bound on degrees of Positivstellensatz calculus proofs for the parity. *Theoret. Comput. Sci.* 259(1–2), 613–622.
 173. Grigoriev, D., & Végh, E. (2012). Complexity of semialgebraic proofs. *Mosc. Math. J.* 12(4), 647–679.
 174. Dechter, R. (2003). *Constraint Processing*. Morgan Kaufmann.
 175. Golub, G. H., & Varga, R. S. (1961). Chebyshev semi-iterative methods, successive over-relaxation iterative methods, and second order Richardson iterative methods. *Numer. Math.* 3, 147–168.
 176. Nesterov, Y. (1983). A method of solving a convex programming problem with convergence rate $O(1/k^2)$. *Soviet Math. Dokl.* 27, 372–376.
 177. Saad, Y. (2003). *Iterative Methods for Sparse Linear Systems* (2nd ed.). SIAM.
 178. Rivlin, T. J. (1990). *Chebyshev Polynomials: From Approximation Theory to Algebra and Number Theory* (2nd ed.). Wiley.
 179. Conley, C. (1978). *Isolated Invariant Sets and the Morse Index*. CBMS Regional Conference Series in Mathematics 38. AMS.
 180. Milnor, J. (1963). *Morse Theory*. Princeton University Press.
 181. Guckenheimer, J., & Holmes, P. (1983). *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields*. Springer.
 182. Putinar, M. (1993). Positive polynomials on compact semi-algebraic sets. *Indiana Univ. Math. J.* 42(3), 969–984.
 183. Stengle, G. (1974). A nullstellensatz and a positivstellensatz in semialgebraic geometry.

- Math. Ann.* 207(2), 87–97.
184. Schmüdgen, K. (1991). The K-moment problem for compact semi-algebraic sets. *Math. Ann.* 289(2), 203–206.
 185. Olah, C., Cammarata, N., Schubert, L., Goh, G., Petrov, M., & Carter, S. (2020). Zoom in: an introduction to circuits. *Distill*.
 186. Elhage, N., Hume, T., Olsson, C., et al. (2022). Toy models of superposition. *Transformer Circuits Thread*.
 187. Choromanska, A., Henaff, M., Mathieu, M., Ben Arous, G., & LeCun, Y. (2015). The loss surfaces of multilayer networks. *AISTATS*.
 188. Li, H., Xu, Z., Taylor, G., Studer, C., & Goldstein, T. (2018). Visualizing the loss landscape of neural nets. *NeurIPS*.
 189. Garipov, T., Izmailov, P., Podoprikin, D., Vetrov, D., & Wilson, A. G. (2018). Loss surfaces, mode connectivity, and fast ensembling of DNNs. *NeurIPS*.
 190. Brea, J., Simsek, B., Illing, B., & Gerstner, W. (2019). Weight-space symmetry in deep networks gives rise to permutation saddles, connected by equal-loss valleys across the loss landscape. arXiv:1907.02911.
 191. Entezari, R., Al-Shedivat, M., Sau, A., & Ribeiro, A. (2022). The role of permutation invariance in linear mode connectivity of neural networks. *ICLR*.
 192. Wong, E., & Kolter, J. Z. (2018). Provable defenses against adversarial examples via the convex outer adversarial polytope. *ICML*.
 193. Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., & Mané, D. (2016). Concrete problems in AI safety. arXiv:1606.06565.
 194. Krakovna, V., Uesato, J., Mikulik, V., Rahtz, M., Everitt, T., Kumar, R., Kenton, Z., Leike, J., & Legg, S. (2020). Specification gaming: the flip side of AI ingenuity. *DeepMind*.